KUSHINGA PHIKELELA POLYTECHNIC

PROJECT TITLE: IMPLEMENTATION OF CITY OF HARARE PROPERTY MANAGEMENT SYSTEM (COH-PMS)

CANDIDATE NUMBER : 0624004G00015

COURSE : INFORMATION TECHNOLOGY

SUBJECT CODE : 553/23/M11

SUBJECT : SOFTWARE PROJECT

SUPERVISOR : MS. MANZINDE

ACADEMIC YEAR : 2024/2026

SOFTWARE PROJECT SUBMITTED IN PARTIAL FULFILLMENT OF NATIONAL DIPLOMA IN INFORMATION TECHNOLOGY

APPROVAL

The undersigned certify that they have supervised the student, research project entitled City of Harare Property Management System, submitted in partial fulfillment of the requirements of the Diploma in Information Technology at Kushinga Phikelela Polytechnic.

Student Signature_____

Date_____

APPROVED FOR EXAMINATION

Supervisor MS. MANZINDE

Signature_____ Date_____

External Examiner_____

Signature_____ Date_____

DECLARATION

I, Warren Mavunga, hereby declare that this project is my original work and has not been submitted to any other college, institution, or university in support of any application for employment, diploma, degree, or any other qualification. I further affirm that this project is being submitted to the Higher Education Examinations Council (HEXCO) in partial fulfillment of the requirements for the award of the National Diploma in Information Technology, and has not been submitted previously for any other examination or qualification.

Signature_____ Date_____

ACKNOWLEDGEMENT

First and foremost, I would like to express my deepest gratitude to Almighty God for the gift of life, strength, wisdom, and guidance throughout the duration of this project. I would like to sincerely acknowledge City of Harare Council for the support rendered during the development of this project. My special appreciation goes to the Mayor and support staff for their valuable contributions in helping me gathers system requirements and better understand the operational challenges faced in property management. I would like to express my heartfelt appreciation to my Project Supervisor, Ms. Manzinde, for her unwavering guidance, patience, constructive criticism, and the time she dedicated to ensuring the success of this project. Your mentorship played a significant role in shaping this work. I would also like to extend my sincere gratitude to my parents for their unwavering support throughout my academic journey. Thank you for your emotional support, encouragement, motivation, and provision of resources that made this project possible. Lastly, to everyone who contributed directly or indirectly to the success of this project, please accept my sincere appreciation.

ABSTRACT

The integration of web technologies in local governance has significantly enhanced administrative efficiency, data transparency, and public service delivery. This project details the development of the City of Harare Property Management System (COH-PMS), a secure web-based application designed to transition municipal venue and property hiring out of error-prone manual methods. The system addresses vulnerabilities in the current system, such as severe delays in checking facility availability, physical paperwork dependency, inconsistent price calculations, and the high risk of property double-bookings.

COH-PMS utilizes strict three-layer architecture to handle municipal workflows seamlessly. The system features specialized front-end dashboards for customers, booking officers, revenue verification officers, and council management. To prevent dual allocations, an automated availability-checking engine verifies venue availability against existing bookings and rechecks before approval or confirmation. Concurrently, a rule-based deterministic pricing framework computes booking fees by applying approved municipal pricing rules, including applicable discounts or adjustments based on venue type, booking period, and level of demand.

The platform was built entirely on an open-source development stack using HTML, CSS, JavaScript, and PHP, backed by a centralized relational MySQL database. Rigorous module and system testing confirmed that the application effectively handles secure multi-role workflows, successfully moving transactions from a customer's initial reservation request through booking officer approval, revenue officer validation (with automated unique receipt generation), and management dashboard updates. Financial and operational feasibility metrics calculated during development projections demonstrate strong system viability, delivering a positive Return on Investment approximately 170% and a short payback period of about 5 months under conservative estimates. Ultimately, COH-PMS aligns with the City of Harare's mission to provide first-class service delivery and promote investment by modernizing real estate governance, sealing revenue leakages, and creating automated, secure audit trails for institutional accountability.

TABLE OF CONTENTS

DECLARATION 3
ACKNOWLEDGEMENT 4
ABSTRACT 5

CHAPTER I: INTRODUCTION 8
1.0 Introduction 8
1.1 Background of the Study 8
1.2 Statement of the Problem 8
1.3 System Objectives 9
1.4 System Justification 9
1.5 Assumptions 9
1.6 Delimitations of the Study 9
1.7 Limitations of the Study 9
1.8 Definition of Terms 10
1.9 Conclusion 10

CHAPTER II: FEASIBILITY STUDY 11
2.0 Introduction 11
2.1 Technical Feasibility 11
2.1.1 Development Tools Stack 11
2.1.2 Administrator Software Requirements 11
2.1.3 Administrator Hardware Requirements 12
2.1.4 Customer Software and Hardware Requirements 12
2.2 Economic Feasibility 13
2.3 Operational Feasibility 14
2.4 Schedule Feasibility 14
2.5 Social Feasibility 15
2.6 Organizational Feasibility 15
2.7 Risk Analysis 16
2.8 Conclusion 16

CHAPTER III: REQUIREMENTS ANALYSIS 17
3.0 Introduction 17
3.1 Current System Analysis 17
3.2 Stakeholder Analysis 22
3.3 Functional Requirements 23
3.4 Non-Functional Requirements 23
3.5 System Inputs 24
3.6 System Processes 24
3.7 System Outputs 25
3.8 System Interface 25
3.9 Conclusion 25

CHAPTER IV: SYSTEM DESIGN 26
4.0 Introduction 26
4.1 System Overview 27
4.2 System Architecture 27
4.3 Use Case Design 30
4.4 System Flowchart Diagram 32
4.5 Sequence Diagram 33
4.6 Database Design 34
4.7 Pricing Rules Design 39
4.8 Interface Design 40
4.9 Security Design 41
4.10 Conclusion 41

CHAPTER V: CODING AND TESTING 42
5.0 Introduction 42
5.1 Programming Languages and Technologies Used 42
5.2 Application File Structure 43
5.3 Application Key Source Codes 44
5.4 Testing 52
5.5 Conclusion 53

CHAPTER VI: IMPLEMENTATION 54
6.0 Introduction 54
6.1 System Deployment 55
6.2 Operational System Interface Demonstration 56
6.3 User Training Framework by Role 61
6.4 Changeover Strategy 61
6.5 Post-Implementation Support 61
6.6 Conclusion 61

CHAPTER VII: SUMMARY, RECOMMENDATIONS AND CONCLUSION 62
7.1 Summary 62
7.2 Recommendations 62
7.3 Recommendations for Further Research 63
7.4 Conclusion 63

CHAPTER VIII: BIBLIOGRAPHY 64

APPENDIX 65

CHAPTER I: INTRODUCTION

1.0 Introduction

This project introduces the City of Harare Property Management System (COH-PMS), a secure web-based system designed to help the City of Harare manage council properties and venues available for hire. The system will provide one central database for authorised council staff to register and update venue details, manage customer booking requests, check venue availability, record payments, apply approved prices, and prevent the same venue from being booked by more than one customer for the same date and time.

Customers, including residents, organisations, and event organisers, will use the system to view available properties and venues, check prices and availability, and submit booking requests. They will follow council-approved payment procedures and provide the relevant payment details or reference for recording and verification by authorised staff. After a booking is approved and the required payment is verified, the system will provide a booking confirmation or receipt to the customer. This will reduce the need for customers to make repeated visits to City of Harare offices, save time and transport costs, and allow them to access venue information more conveniently.

COH-PMS will also use approved pricing rules to adjust property or venue charges according to the type of venue, purpose of hire, booking period, or level of demand. The system will generate reports on facility usage and revenue so that management can monitor bookings, payments, frequently used venues, and income collected. This will support better record keeping, improve service delivery for customers, reduce manual work for council staff, and assist management in making decisions about council properties and venues.

1.1 Background of the Study

Public properties are important resources for local authorities because they provide places for community activities and help councils collect revenue. In Harare, facilities such as halls, stadia, and community centres are used for meetings, sports events, weddings, and other public or private activities. These facilities need proper management so that they remain available, are used correctly, and continue to serve the needs of residents and organisations.

Managing property and venue hire requires correct information about facilities, available dates, customers, bookings, prices, payments, and revenue. It also requires that the same venue is not booked by more than one customer at the same time. The City of Harare manages public facilities that residents and organisations hire for events and activities, including community halls and centres in areas such as Highfield, Mbare, and Glen View, as well as stadia such as Rufaro, Gwanzura, and Warren Park.

At present, this work is done using paper records, physical forms, manual booking registers, and office visits. Customers travel to council offices to check availability, complete forms, follow council-approved payment procedures, and confirm bookings. They may need to visit more than once before a booking is finalised, which takes time and increases transport costs. Staff may also find it difficult to check bookings, update cancellations, confirm payments, and prepare reports. Mistakes may occur when records are written incorrectly, misplaced, or not updated in time, and the same venue can be booked twice.

Separate records in different offices can also make it difficult for management to see how venues are used and how much revenue is collected. The City of Harare tariff schedule shows different charges for different uses and booking periods. Manual calculation can produce incorrect or inconsistent charges and make it difficult to apply approved lower prices during quieter periods. Evidence on local government revenue shows that weak billing and late collection can cause revenue losses, while ICT tools such as online payment systems and digital records can improve collection and transparency.

1.2 Statement of the Problem

The City of Harare currently manages property and venue hire through manual processes such as paper booking registers, physical forms, office visits, and separate records. These methods cause delays in checking venue availability, making bookings, confirming payments, and preparing reports. They may also result in incorrect or missing records, double-booking of venues, inconsistent property prices, and limited information on facility usage and revenue. Therefore, there is a need for a secure web-based City of Harare Property Management System to manage venues, automatically adjust property prices using approved rules, and provide reports on facility usage and revenue.

1.3 System Objectives

The system objectives are:

- To implement a secure web-based venue management system.
- To automatically adjust property prices using approved rules.
- To produce reports on facility usage and revenue.

1.4 System Justification

- The proposed system will allow customers to view available venues, submit booking requests, follow council-approved payment procedures, and receive booking confirmation without making repeated visits to council offices.
- It will centralise property, customer, booking, and payment information, improve access to venue availability, reduce the risk of double-booking, and support approved automated pricing rules.
- It will provide timely reports to help management monitor facility usage and revenue, make informed decisions, and improve service delivery to residents and organisations.

1.5 Assumptions

- The study assumes that authorised City of Harare staff have basic computer skills and will be able to use the COH-PMS after simple training.
- It assumes that the council will provide correct property, venue, price, booking, and payment information for use in the system.
- It assumes that council staff and customers will have access to internet-enabled devices when using the web-based system.

1.6 Delimitations of the Study

This study focuses on the development of the City of Harare Property Management System (COH-PMS) for managing council properties and venues available for hire. The system will cover property and venue records, venue availability, booking requests, approved pricing rules, payment recording and verification, booking confirmations, receipts, and reports on facility usage and revenue. The system will be limited to the management of properties and venues available for hire. It will not cover property maintenance, property sales, staff salaries, or other council services.

1.7 Limitations of the Study

The study may be limited by time, limited access to some council records, limited user feedback, and internet or computer problems during system testing. These limitations will be reduced by using available records, consulting selected users, and testing the system with sample data where necessary.

1.8 Definition of Terms

- Property Management System: A computerised system used to manage information about properties, venues, customers, bookings, prices, payments, and reports.
- Venue: A public place or facility that can be hired for an event, such as a hall, stadium, or community centre.
- Customer: A person or organisation that hires a City of Harare property or venue.
- Booking: The process of reserving a venue for use on a selected date and time.
- Booking Request: A request submitted by a customer to hire a selected venue for a chosen date and time.
- Booking Confirmation: A notice or receipt given to a customer after a booking has been approved and payment has been confirmed.
- Double-booking: A situation in which the same venue is reserved for more than one customer for the same date and time, or for overlapping periods.
- Web-based System: A system that is accessed through a web browser using an internet-enabled device.
- Price Adjustment: The change of a property or venue price according to approved rules, such as venue type, booking period, or level of demand.
- Approved Rules: Pricing and booking conditions formally authorised by the City of Harare, which the system applies automatically to determine charges, discounts, and booking eligibility.
- Facility Usage: Information showing how often a venue or property is booked and used.
- Revenue: Money collected by the City of Harare from property and venue hire.
- Report: A document or screen display that shows organised information, such as bookings, payments, facility usage, and revenue.

1.9 Conclusion

This chapter introduced the City of Harare Property Management System. It explained the background of the study, the problem being addressed, the objectives, justification, assumptions, delimitations, limitations, and key terms. The chapter showed that manual property and venue management can cause delays, booking errors, pricing problems, and reporting challenges for both customers and council staff. The next chapter evaluates the feasibility of the proposed COH-PMS across technical, economic, operational, organisational, schedule, and social dimensions.

CHAPTER II: FEASIBILITY STUDY

2.0 Introduction

This chapter examines the feasibility of the proposed City of Harare Property Management System (COH-PMS) to determine whether the proposed system can be developed, implemented, and used successfully within the City of Harare environment. It assesses whether the system is technically possible, affordable, practical for council staff and customers, and suitable for the organisation. It considers the hardware, software, internet connection, staff skills, cost, expected benefits, security, and time needed to develop and implement the system. The purpose is to determine whether COH-PMS is a suitable solution for improving property and venue management, customer booking, payment recording, price adjustment, and reporting on facility usage and revenue.

2.1 Technical Feasibility

Technical feasibility examines the technical skills and requirements to develop and use the City of Harare Property Management System (COH-PMS). The proposed system will be developed as a web-based application using affordable and widely available technologies.

Table 2.1.2 Administrator Software Requirements

Software	Purpose
Windows 10 or Later	Operating system for system development and testing
Apache Server	Runs the PHP web application locally
PHP	Processes logins, bookings, prices, payments, and reports
MySQL	Stores user, venue, booking, pricing-rule, and payment records; reports are generated from stored data
Web Browser	Allows staff and customers to access the system

2.1.3 Administrator Hardware Requirements

Requirement	Minimum Requirement	Purpose
Processor	Dual-core processor	Used to manage venues, bookings, prices, payments, and reports
Storage	50 GB free space	Stores system files, records, and backups
RAM	8 GB RAM	Speed data processing
Printer	Optional	Prints booking confirmations, receipts, and reports

2.1.4 Customer Software and Hardware Requirements

Requirement	Minimum Requirement	Purpose
Smartphone, Tablet, or Computer	Internet-enabled device	Views venues, checks availability, submits booking requests, and accesses booking/payment information
Internet connection	Mobile data or Wi-Fi connection	Accesses the system online
Web browser	Google Chrome, Microsoft Edge, Firefox, or mobile browser	Opens and uses the customer system pages
Email Address or Phone Number	Valid contact details	Receives booking updates and payment confirmation

The listed tools, software, and hardware are affordable and widely available. Therefore, the proposed system is technically feasible.

2.2 Economic Feasibility

Economic feasibility examines whether the expected benefits of the City of Harare Property Management System (COH-PMS) are worth the costs of developing and operating it. The following table compares the estimated costs and benefits of the proposed system.

Table 2.2.1 Cost-Benefit Analysis

Category	Type	Item	Estimated Value ($)
Cost	Non-recurring	Internet data and research	50.00
Cost	Non-recurring	Transport for data collection and meetings	60.00
Cost	Non-recurring	System testing and basic staff orientation	50.00
Cost	Non-recurring	Printing, binding and stationery	40.00
Cost	Recurring	Domain registration and shared web hosting (Year 1)	70.00
		Total Estimated Costs (Year 1)	270.00
Benefit	Annual	Reduced paper, printing and physical form costs	80.00
Benefit	Annual	Staff time savings for availability checks, booking updates and report preparation	250.00
Benefit	Annual	Reduced double-booking and improved payment recording and revenue monitoring	400.00
		Total Estimated Annual Benefits	730.00

Net Annual Benefit = Total Annual Benefits − Recurring Costs
= $730 − $70
= $660 per year

Return on Investment (ROI) = [(Total Annual Benefits − Total Costs) / Total Costs] × 100
= [($730 − $270) / $270] × 100
= 170%

Payback Period = Initial Investment / Total Annual Benefits
= $270 / $660
= 0.41 years (Approximately 5 months)

2.3 Operational Feasibility

Operational feasibility evaluates whether the City of Harare Property Management System (COH-PMS) can fit into the daily work of council staff and the venue-hiring process used by customers.

- Primary Users: Customers, Booking Officers, Revenue Officers, System Administrators, and Council Management.
- Customer Activities: Customers will view available venues, check prices and availability, submit booking requests, follow council-approved payment procedures, and receive booking confirmations or receipts.
- Booking Officer Activities: Booking officers will manage venue details, review booking requests, approve or reject bookings, and update booking status. Revenue Officers will record and verify payments.
- Administrator Activities: Administrators will manage user accounts, approved pricing rules, and reports on facility usage and revenue.
- Workflow Integration: The system will replace paper booking registers and physical forms with one central database. Booking Officers will approve or reject requests, while Revenue Officers will record and verify payments. A booking will be confirmed only after the required approval and payment verification, allowing the City of Harare to remain in control of property and venue hire.
- Training Requirements: Basic training will be provided to council staff on managing venues, bookings, payments, price adjustments, and reports. Customers will need little training because the system will use clear forms and simple instructions.

The proposed COH-PMS is operationally feasible because it supports the daily work of council staff while providing customers with a simpler way to access venue information, submit bookings, follow the approved payment process, and receive confirmation.

2.4 Schedule Feasibility

Schedule feasibility assesses whether the City of Harare Property Management System (COH-PMS) can be developed, tested, and deployed within the available project period. The project spans eight months from February 2026 to September 2026, divided into manageable development phases.

Table 2.4.1 Project Gantt Chart

Task	February	March	April	May	June	July	August	September
Project Proposal								
Planning								
Analysis								
Design								
Coding & Testing								
Implementation								
Documentation								

2.5 Social Feasibility

The proposed COH-PMS is socially feasible because it will improve access to City of Harare properties and venues for residents, organisations, and event organisers.

- Improved Public Access: Customers will be able to view venue information, availability, prices, and booking requirements without relying entirely on physical visits.
- Reduced Inconvenience: The system will reduce long queues, unnecessary travelling, paperwork, and delays when making venue-hire requests.
- Inclusive Technology: The system will use a simple web interface that can be accessed by users with basic computer or smartphone skills.
- Improved Transparency: Customers will receive clearer information about bookings, payments, prices, and confirmations, reducing misunderstandings and disputes.
- Data Privacy: Customer, booking, and payment information will be protected through controlled access and appropriate security measures.

2.6 Organisational Feasibility

The proposed COH-PMS is organisationally feasible because it supports the City of Harare's responsibility to manage public properties efficiently and improve service delivery, in the following ways:

- Improved Property Management: The system will centralise information about council properties, venues, bookings, prices, and payments.
- Institutional Efficiency: Council staff will spend less time searching through paper records, checking availability manually, and preparing reports.
- Revenue Management: The system will support the consistent application of approved pricing rules and improve the recording and monitoring of venue-hire revenue.
- Management Decision-Making: Reports on facility usage, bookings, payments, and revenue will help council management make informed decisions.
- Organisational Alignment: The system supports the council's objectives of improving public services, reducing administrative errors, increasing accountability, and making better use of municipal facilities.

2.7 Risk Analysis

Risk analysis identifies possible risks that may affect the successful development and use of the City of Harare Property Management System (COH-PMS), together with measures for reducing their effects.

Table 2.7.1 Risk and Effects Analysis

Risk	Impact	Mitigation Strategy
System downtime	High	Regular system maintenance, backups, and a reliable hosting environment will be used.
Internet connection failure	Medium	Staff will use stable internet connections, and interrupted transactions will be recorded for follow-up.
Incorrect booking information	Medium	Input validation, clear forms, and staff review of booking requests will reduce errors.
Double-booking	High	The system will check venue availability and prevent conflicting bookings for the same date and time.
Unauthorised access	High	User accounts, passwords, and role-based access control will protect system information.
Data loss	High	Regular database backups and recovery procedures will be maintained.
User resistance	Medium	Staff training and a simple user interface will support adoption of the system.
Incorrect price adjustments	Medium	Approved pricing rules will be reviewed and updated only by authorised administrators.

2.8 Conclusion

This chapter established that the proposed City of Harare Property Management System (COH-PMS) is technically, economically, operationally, socially, and organisationally feasible. The system can be developed using affordable and widely available web technologies. The economic analysis showed a positive return and a short payback period even under conservative assumptions. Operationally, the system fits the daily work of Booking Officers, Revenue Officers, Administrators, and Council Management while giving customers a simpler way to view venues, submit booking requests, and receive confirmations. Socially, it improves public access, reduces travelling and paperwork, and increases transparency. Organisationally, it supports the City of Harare's goals of better property management, consistent pricing, improved revenue monitoring, and informed decision-making. Potential risks such as system downtime, data loss, unauthorised access, and double-booking can be managed through regular backups, role-based access control, system monitoring, and automated availability checks. Therefore, the COH-PMS is recommended for development and implementation.

CHAPTER III: REQUIREMENTS ANALYSIS

3.0 Introduction

This chapter presents the requirements analysis for the proposed City of Harare Property Management System (COH-PMS). The purpose of this phase is to define the system requirements in detail, including the current system, stakeholders, functional requirements, non-functional requirements, system inputs, processes, outputs, and interfaces. Requirements analysis identifies what the system must do to address the challenges of manual property and venue management. It considers the needs of customers, booking officers, administrators, and council management. The requirements identified in this chapter will guide the design, development, and testing of the COH-PMS.

3.1 Current System Analysis

The current City of Harare property and venue-hire process is mainly manual:

- Property enquiry: Customer visits a City of Harare office to check venue availability.
- Manual check: Booking officer reviews a paper register to confirm availability.
- Customer briefing: Officer details the charges, dates, and hiring conditions.
- Form completion: Customer fills out a physical booking form with personal or organizational details.
- Request logging: Booking officer enters the request into a manual register or file.
- Payment processing: Customer pays at the council's cashier or payment office.
- Booking confirmation: Officer manually records the payment and confirms the booking.
- Receipt issuance: Customer receives a physical receipt or confirmation document.
- Manual reporting: Staff compiles reports on revenue and venue usage by reviewing paper files.

Table 3.1.1 Current System Analysis

Question	Current System
What is being done?	Council properties and venues are hired to customers for meetings, events, sports, weddings, and other activities.
How is it being done?	Customers visit council offices, check paper registers, complete physical forms, make payments, and receive confirmation from staff.
Who is doing it?	Customers, booking officers, cashiers or revenue officers, and council administrators.
When is it being done?	When customers need to enquire about, book, pay for, cancel, or confirm the hire of a property or venue.
Why is it being done?	To allocate council venues to customers and collect revenue from property and venue hire.
How can it be improved?	By introducing the COH-PMS to centralise venue records, show availability, manage bookings and payments, apply approved pricing rules, prevent double-booking, and generate reports.

Figure 3.1.2 Level 0 DFD Diagram – Existing Manual System at City Of Harare Council

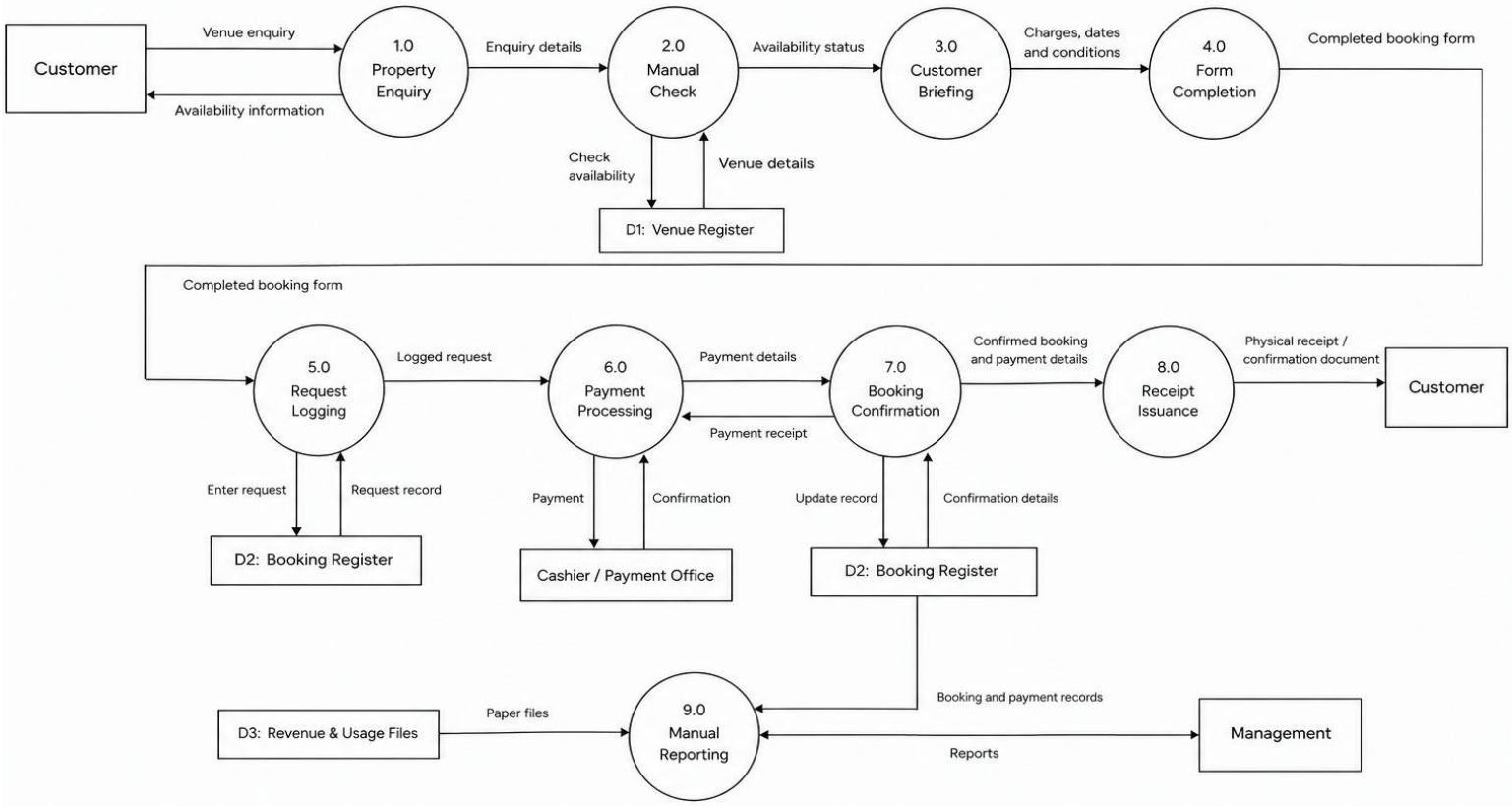


Figure 3.1.2 Level 0 DFD Diagram – Existing Manual System at City Of Harare Council

Figure 3.1.3 UML Use Case Diagram – Existing Manual System at City Of Harare Council

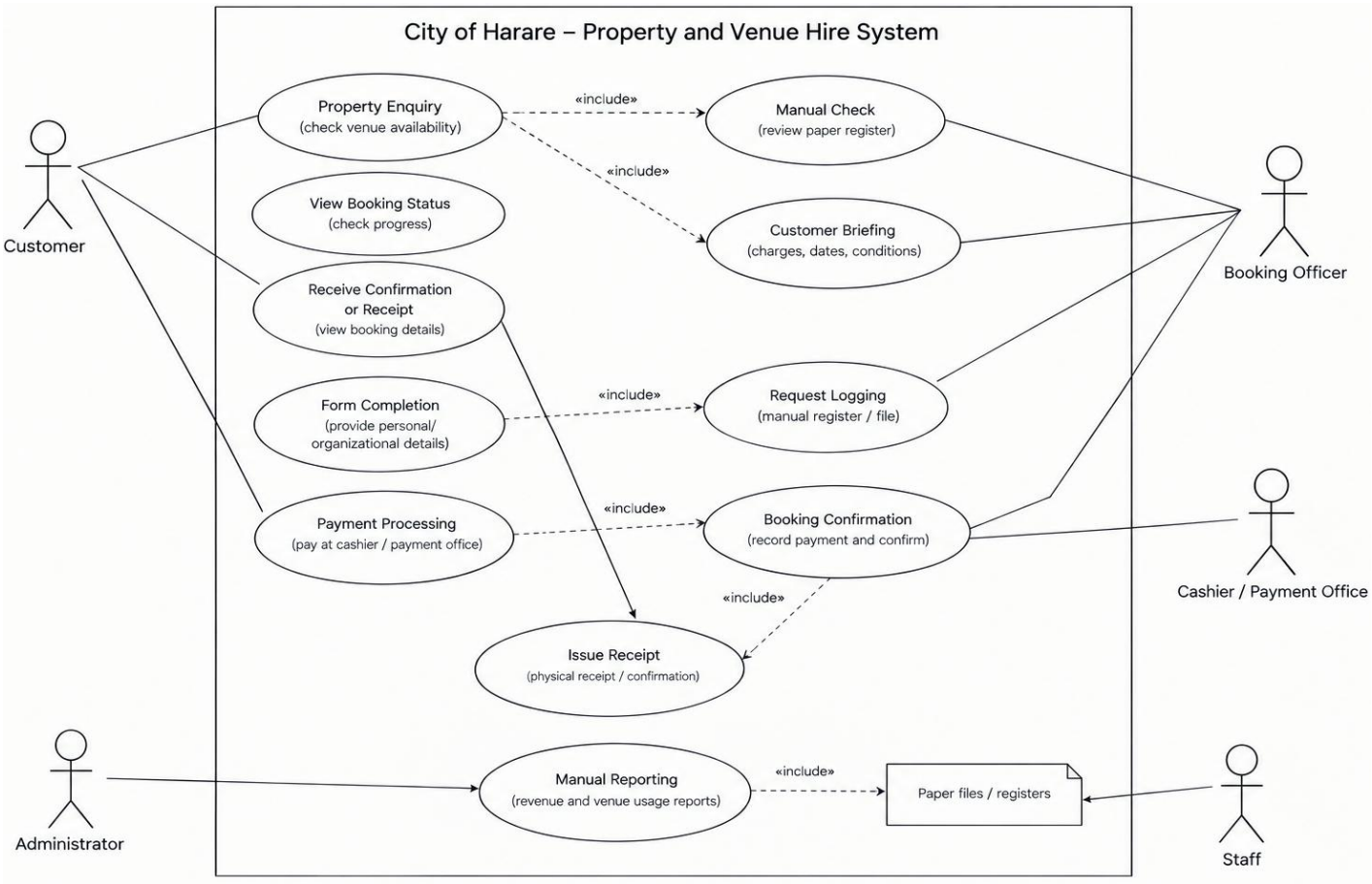


Figure 3.1.3 UML Use Case Diagram – Existing Manual System at City Of Harare Council

Figure 3.1.4 Flowchart Diagram – Existing Manual System at City Of Harare Council

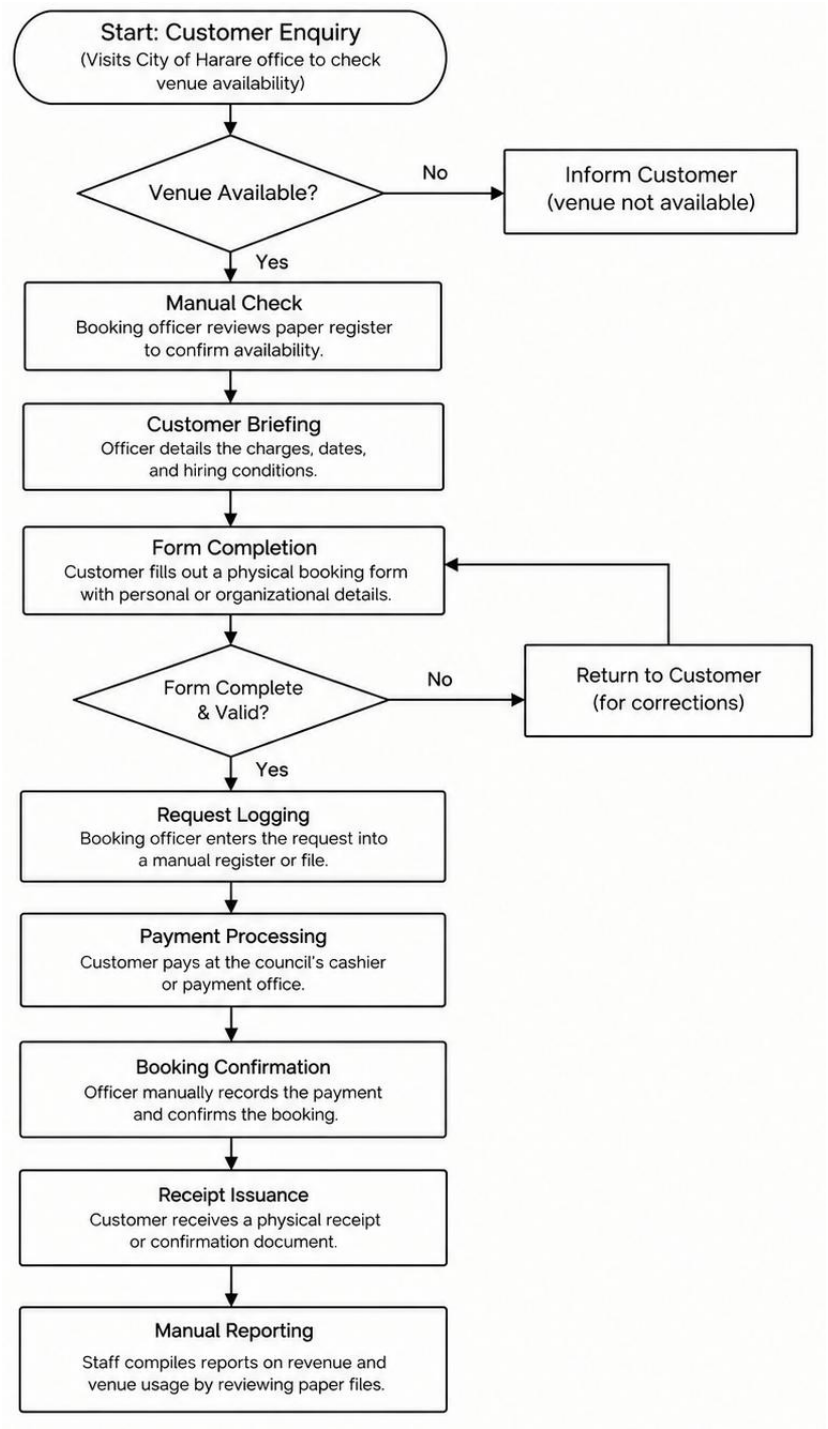


Figure 3.1.4 Flowchart Diagram – Existing Manual System at City Of Harare Council

Figure 3.1.5 UML Sequence Diagram – Existing Manual System at City Of Harare Council

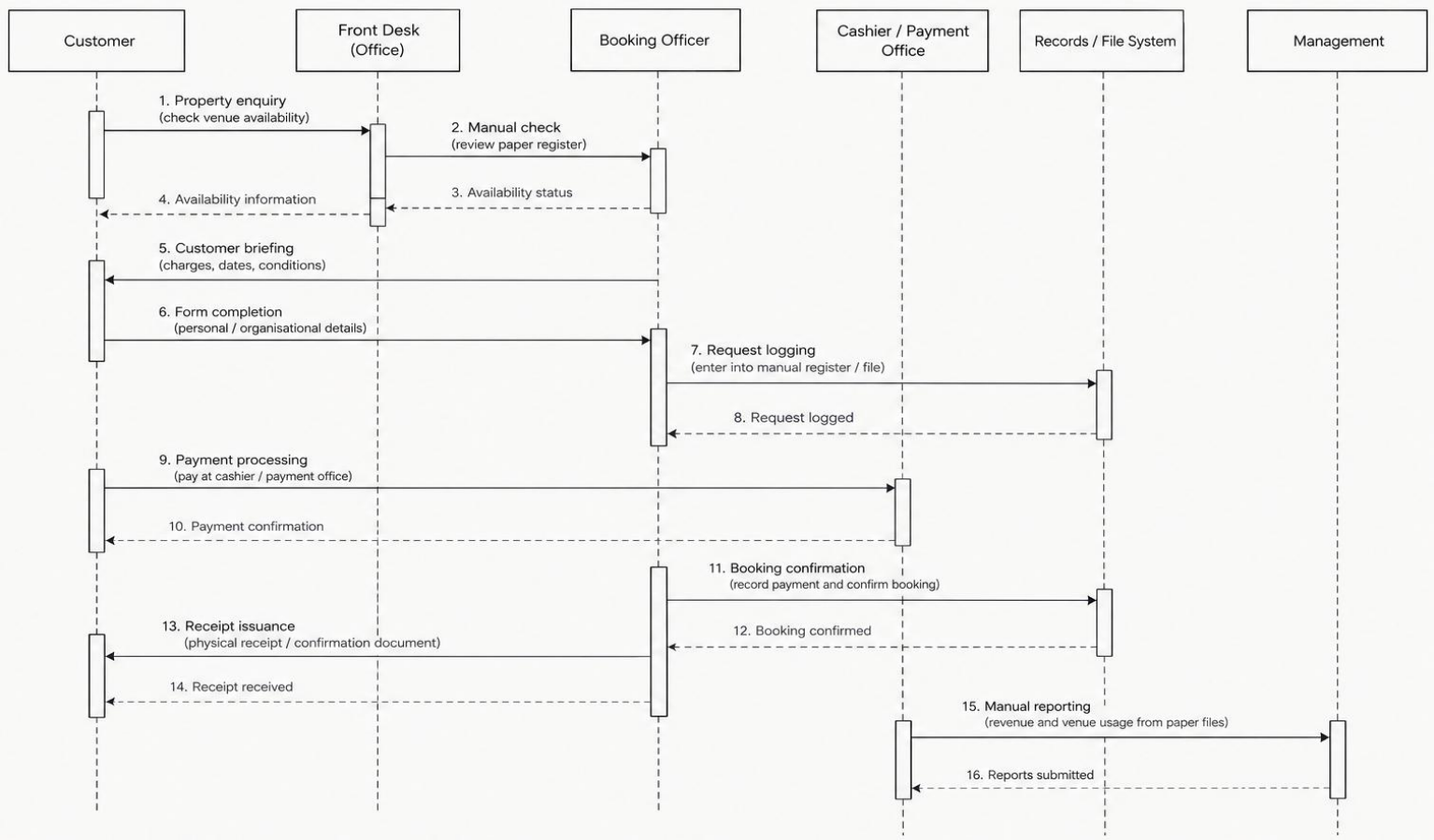


Figure 3.1.5 UML Sequence Diagram – Existing Manual System at City Of Harare Council

Problems with the current manual system:

- Booking delays: Customers must visit Council offices to make enquiries, complete forms, and submit payments, resulting in long waiting times and queues.
- Double booking: Venue availability is checked manually, increasing the risk of booking the same venue for overlapping dates and times.
- Poor record keeping: Paper records can be misplaced, damaged, incomplete, or difficult to retrieve.
- Inconsistent pricing: Manual calculation of charges may lead to inconsistent application of discounts, surcharges, and approved pricing rules.
- Delayed payment verification: Manual recording and verification of payments can delay booking confirmation and make payment tracking difficult.
- Difficult report preparation: Booking, payment, facility-usage, and revenue reports are prepared manually, which is time-consuming and prone to errors.
- Limited management visibility: Management has limited access to current booking and revenue information stored in separate paper records.
- Limited customer convenience: Customers cannot easily check availability, submit bookings, or track booking status remotely.

3.2 Stakeholder Analysis

Stakeholder analysis identifies the people and groups who will use manages, benefit from, or support the City of Harare Property Management System (COH-PMS). Their needs were considered when defining the system requirements.

Table 3.2.1 Stakeholder Analysis

Stakeholder	Role	Key Needs	System Interaction
Customers	Primary users	Venue information, availability, correct prices, simple booking, payment confirmation	View venues, check availability, submit booking requests, make payments, and receive confirmations
Booking Officers	Operational users	Updated booking records, easy approval process, reduced paperwork	Manage venue details, review booking requests, approve or reject bookings, and update booking status
Cashiers / Revenue Officers	Payment users	Accurate payment records and easy payment verification	Record payments, verify transactions, and issue receipts
System Administrators	System managers	Secure access, user management, correct pricing rules, and system control	Manage user accounts, venue records, pricing rules, and system settings
Council Management	Decision makers	Accurate reports on bookings, facility usage, payments, and revenue	View reports and use information for monitoring and decision-making
City of Harare	System owner / beneficiary	Improved service delivery, accountability, revenue collection, and property management	Provides property information, policies, approved pricing rules, and staff support
Developer	System builder	Clear requirements, user feedback, and a testable system	Designs, develops, tests, and maintains the COH-PMS

3.3 Functional Requirements

Functional requirements describe the services and functions that the City of Harare Property Management System (COH-PMS) must perform. The system must be able to:

- Allow customers to register, log in, and manage their accounts.
- Display available council properties and venues with their details, location, capacity, prices, and booking conditions.
- Allow customers to search for venues and check availability for selected dates and times.
- Allow customers to submit booking requests for selected properties or venues.
- Prevent double-booking by checking whether a venue is already booked for the selected date and time. The system shall recheck availability before approval or confirmation. Pending requests shall not permanently reserve a time slot; the system shall apply any temporary hold only if a hold period is formally defined.
- Calculate venue-hire charges using approved pricing rules, including applicable discounts or price adjustments.
- Allow booking officers to view, approve, reject, cancel, or update booking requests.
- Allow authorised Revenue Officers to record and verify payments against the relevant booking.
- Generate booking confirmations and payment receipts for customers.
- Allow administrators to manage user accounts, property records, venue details, and approved pricing rules.
- Generate reports on bookings, payments, facility usage, and revenue.
- Allow authorised users to log out securely.

3.4 Non-Functional Requirements

Non-functional requirements describe how the COH-PMS should perform its functions and the standards it must meet.

- Usability: The system should be simple, clear, and easy to use for customers and council staff with basic computer skills.
- Performance: The system should display venue information, availability results, bookings, and reports within a reasonable time.
- Security: The system should require usernames and passwords and use role-based access to protect customer, booking, payment, and administrative information.
- Reliability: The system should save valid records correctly and prevent duplicate or conflicting venue bookings.
- Availability: The web-based system should be accessible to authorised users whenever an internet connection is available.
- Data Integrity: The system should validate required fields, prevent incorrect data entry, and maintain accurate records for venues, bookings, prices, and payments.
- Backup and Recovery: The database should be backed up regularly so that information can be recovered after system failure or data loss.
- Compatibility: The system should operate through common web browsers on computers, tablets, and smartphones.

- Maintainability: Authorised administrators should be able to update venue details, user accounts, and approved pricing rules when required.

- Scalability: The system should support the addition of more council properties, venues, users, bookings, and reports in the future.

3.5 System Inputs

The inputs to the City of Harare Property Management System (COH-PMS) include the following:

- Customer Details: Customer name, contact number, email address, address, organisation name where applicable, username, and password.

- Property and Venue Details: Venue name, property type, location, capacity, facilities available, description, hire price, and availability status.

- Booking Details: Selected venue, booking date, start time, end time, event type, expected number of people, and booking purpose.

- Pricing Details: Standard venue price, applicable approved discounts, booking period, demand level, and final amount payable.

- Payment Details: Booking reference number, amount paid, payment date, payment method, transaction reference where applicable, and payment status.
- Report Details: Selected report type, date range, booking status, payment status, facility usage, and revenue information.

3.6 System Processes

The COH-PMS will process the information entered by customers and council staff through the following activities:

- User Registration and Login: The system captures customer and staff details, creates user accounts, and verifies login credentials before allowing access.

- Venue Management: Administrators or booking officers add, update, view, or remove property and venue records.

- Availability Checking: The system checks existing bookings to determine whether a selected venue is available for the requested date and time.

- Booking Management: Customers submit booking requests, while booking officers review, approve, reject, cancel, or update bookings.

- Pricing Calculation: The system applies approved pricing rules to calculate the correct hire charge, including applicable price adjustments or discounts.

- Payment Recording: Authorised Revenue Officers record payment details against the relevant booking and verify the payment before confirmation.

- Confirmation Generation: The system generates a booking confirmation after the booking has been approved and the required payment has been verified. A payment receipt is generated for the recorded payment.

- Report Generation: The system processes stored data to produce reports on bookings, venue usage, payments, and revenue.

3.7 System Outputs

The COH-PMS will produce the following outputs:

- Venue Information: Details of council properties and venues, including location, capacity, facilities, price, and availability.
- Availability Results: A message showing whether the selected venue is available or unavailable for the requested date and time.
- Booking Status: Booking request status, such as pending, approved, rejected, cancelled, or confirmed.
- Calculated Hire Charge: The total amount payable after the system applies approved pricing rules and any applicable adjustments.
- Booking Confirmation: A confirmation document or screen showing the customer, venue, booking date, time, event details, and booking reference number.
- Payment Receipt: A receipt showing the booking reference, amount paid, payment date, transaction reference, and payment status.
- Management Reports: Reports on property bookings, venue usage, payments received, outstanding payments, and revenue collected.

3.8 System Interface

The COH-PMS will use the following interfaces to allow users to interact with the system:

- User Interface: A web-based interface with pages for registration, login, venue viewing, availability checking, booking requests, payment recording, booking management, pricing management, and report generation.
- Customer Interface: Customers will use a browser to create accounts, view venue details, check availability, submit booking requests, and view confirmations.
- Staff Interface: Booking officers, revenue officers, and administrators will use dashboards to manage properties, bookings, payments, user accounts, pricing rules, and reports.
- Database Interface: The system will use a central database to store user, venue, booking, pricing-rule, and payment information. Reports will be generated from these records rather than stored as a separate core record type.
- Payment Interface: The system will record payment information and support staff verification of payments. Integration with an external payment service is optional and will only be included if an approved service and the required access are available; it is not required for the core system.
- Hardware Interface: Users will access the system through internet-enabled computers, tablets, or smartphones.

3.9 Conclusion

This chapter presented the requirements analysis for the City of Harare Property Management System (COH-PMS). It examined the current manual property and venue-hire process, identified the stakeholders, and defined the functional and non-functional requirements of the proposed system. The analysis established that the COH-PMS must manage venue information, availability, bookings, approved price adjustments, payments, confirmations, and reports. These requirements will guide the design and development of the system in the next chapter.

CHAPTER IV: SYSTEM DESIGN

4.0 Introduction

This chapter presents the system design for the proposed City of Harare Property Management System (COH-PMS). The design translates the requirements identified in Chapter 3 into a clear structure for the web-based system. It describes how the system will manage customer accounts, properties and venues, bookings, approved pricing rules, payments, confirmations, and reports on facility usage and revenue. The chapter covers the system architecture, database design, user interfaces, and security controls that will guide the development of the COH-PMS.

4.1 System Overview

The City of Harare Property Management System (COH-PMS) is designed as a secure web-based system for managing council properties and venues available for hire. The system will replace manual booking registers and paper forms with one central database that stores venue, customer, booking, pricing, payment, and report information.

The design is based on the following principles:

- Centralised Management: All property, venue, booking, payment, and pricing information will be stored in one central system.
- User-Based Access: Customers, booking officers, revenue officers, administrators, and council management will access functions according to their roles.
- Availability Control: The system will check venue availability before a booking is submitted or approved to prevent double-booking.
- Approved Pricing Rules: The system will calculate venue-hire charges using pricing rules approved by the City of Harare.
- Payment and Confirmation: Payments will be recorded against bookings, and confirmations or receipts will be generated after payment verification.
- Reporting: The system will produce reports on bookings, facility usage, payments, and revenue to support council management decisions.

4.2 System Architecture

The City of Harare Property Management System (COH-PMS) follows a three-layer web-based architecture consisting of the Presentation, Application, and Data layers. The architecture supports council property and venue management, customer bookings, approved pricing rules, payments, and management reports.

Presentation Layer

- Web Interface: Accessed through a browser on computers, tablets, or smartphones.
- Role-Based Dashboards:
    i. Customers: Register, log in, view venue information, check availability, submit booking requests, and view confirmations or receipts.
    ii. Booking Officers: Use the staff dashboard to manage venue details and booking requests.
    iii. Revenue Officers: Record and verify customer payments.
    iv. Administrators: Manage system users, venues, and approved pricing rules.
    v. Council Management: Access reports on facility usage and revenue.

Application Layer

- Authentication and Access Control: Manages user login, authentication, and role-based access to system functions.
- Core Business Modules:
    i. Venue Management Module: Manages council property and venue records.
    ii. Booking Management Module: Records booking requests and updates their status as pending, approved, rejected, cancelled, or confirmed. A booking is confirmed only after the required approval and payment verification.
    iii. Availability-Checking Module: Checks a requested venue, date, and time against existing bookings and rechecks availability before approval or confirmation to prevent overlapping confirmed bookings.
    iv. Pricing Calculation Module: Calculates hire charges by applying approved property-price rules.
    v. Payment Module: Records and verifies payments, then links them to the relevant bookings.
    vi. Confirmation and Receipt Module: Generates booking confirmations and payment receipts.
    vii. Reporting Module: Produces reports on bookings, facility usage, payments, and revenue.

Data Layer

- Central Database: Stores all system information used by the COH-PMS.
- Database Records:
    i. Users Record: Stores account details, securely hashed passwords, and assigned user roles.
    ii. Venues Record: Stores property information, including name, location, capacity, facilities, price, and availability status.
    iii. Bookings Record: Stores booking references, customer and venue references, dates, times, event details, calculated charges, and booking status.
    iv. Pricing Rules Record: Stores approved conditions for price adjustments or discounts.
    v. Payments Record: Stores payment references, booking references, amounts, payment methods, transaction references, dates, and payment status.
- Report Generation: Management reports are generated from the venue, booking, and payment records stored in the central database.

4.2.1 Summarized Table of the System Architecture

Layer	Content
Presentation Layer	Customer, Booking Officer, Revenue Officer, Administrator, and Council Management web interfaces.
Application Layer	Authentication and access control; venue management; booking management; availability checking; approved pricing; payment recording and verification; confirmation and receipt generation; reporting.
Data Layer	Central MySQL database containing Users, Venues, Bookings, Pricing Rules, and Payments. Reports are generated from stored records.

Figure 4.2.2 System Architecture Layout of COH-PMS

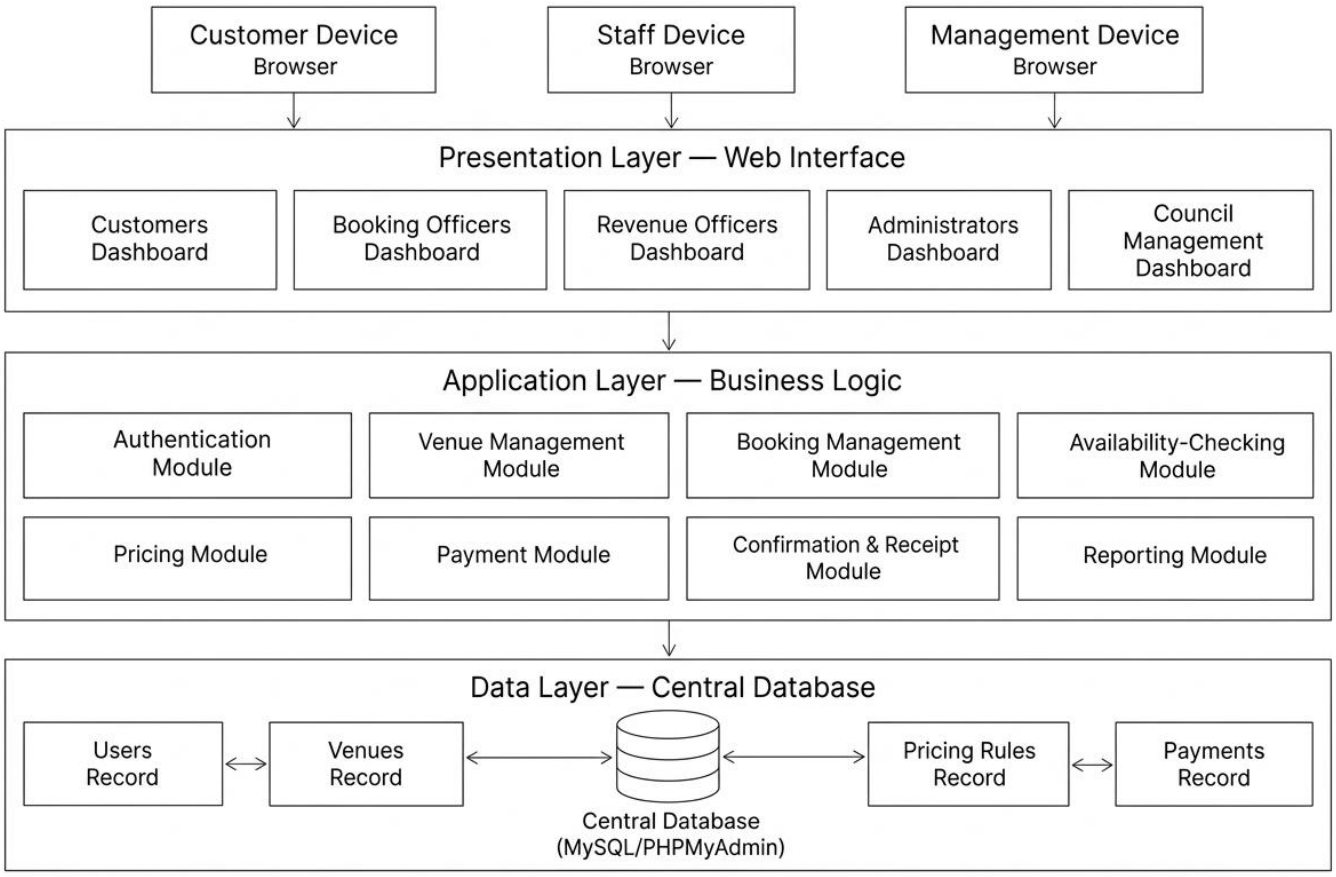


Figure 4.2.2 System Architecture Layout of COH-PMS

The Level 0 Data Flow Diagram for the proposed City of Harare Property Management System (COH-PMS) shows how system users interact with the centralised web-based system and database. A customer submits a booking request, the system checks venue availability and calculates the booking charge, the Booking Officer approves or rejects the booking, and the customer submits payment details. The Revenue Officer verifies the payment and confirms the booking, while Council Management accesses facility-usage and revenue reports generated from the system database.

Figure 4.2.2 Level 0 DFD Diagram of COH-PMS

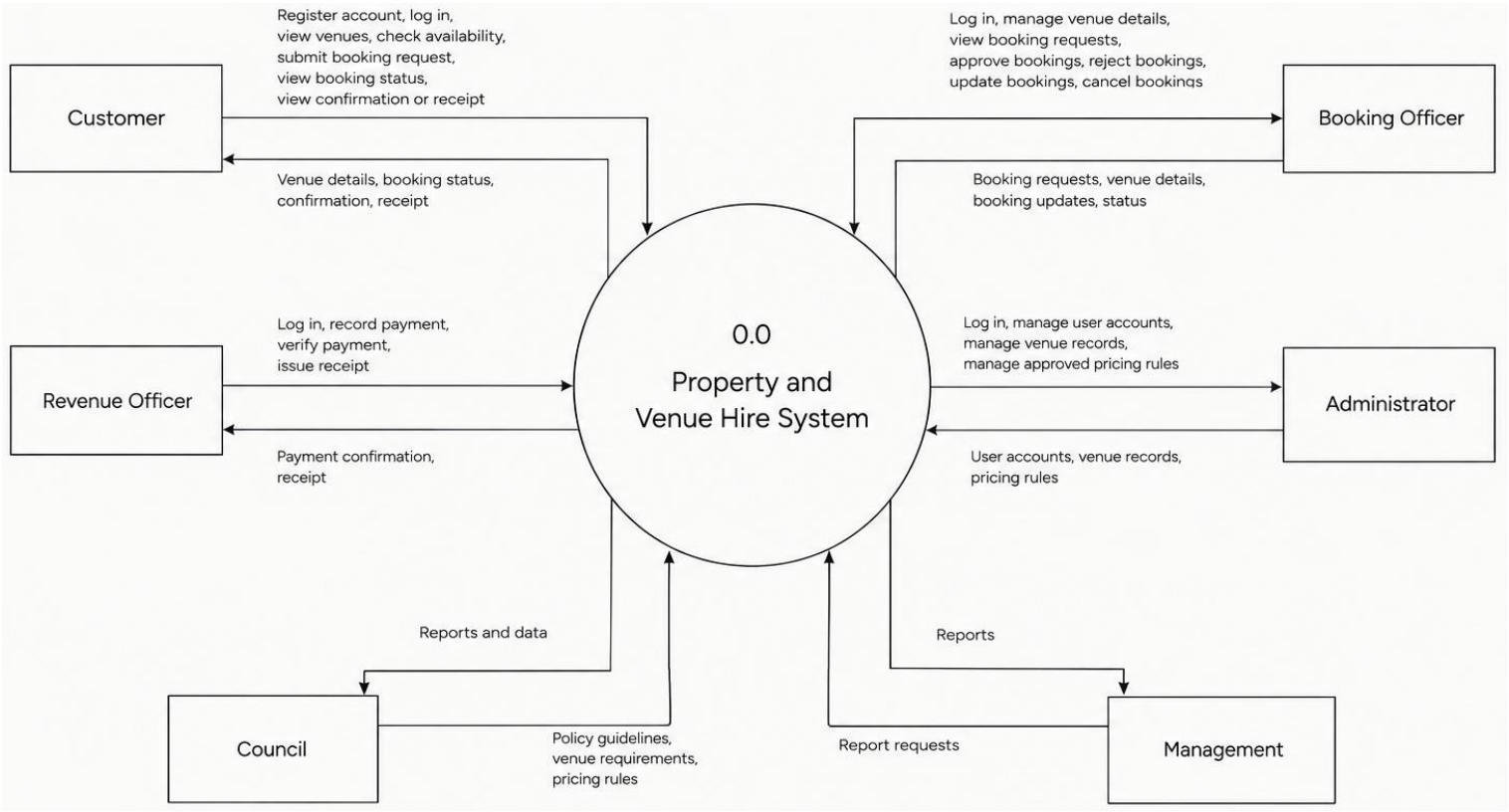


Figure 4.2.2 Level 0 DFD Diagram of COH-PMS

4.3 Use Case Design

The use case design shows how different users interact with the City of Harare Property Management System (COH-PMS). It identifies the main system functions available to customers, council staff, administrators, and management.

Table 4.3.1 Use Case Summary

Actor	Use Cases
Customer	Register account, log in, view venues, check availability, submit booking request, view booking status, view confirmation or receipt
Booking Officer	Log in, manage venue details, view booking requests, approve bookings, reject bookings, update bookings, cancel bookings
Revenue Officer	Log in, record payment, verify payment, issue receipt
Administrator	Log in, manage user accounts, manage venue records, manage approved pricing rules
Council Management	Log in, view facility-usage reports, view payment reports, view revenue reports

Figure 4.3.2 UML Use Case Diagram – COH-PMS

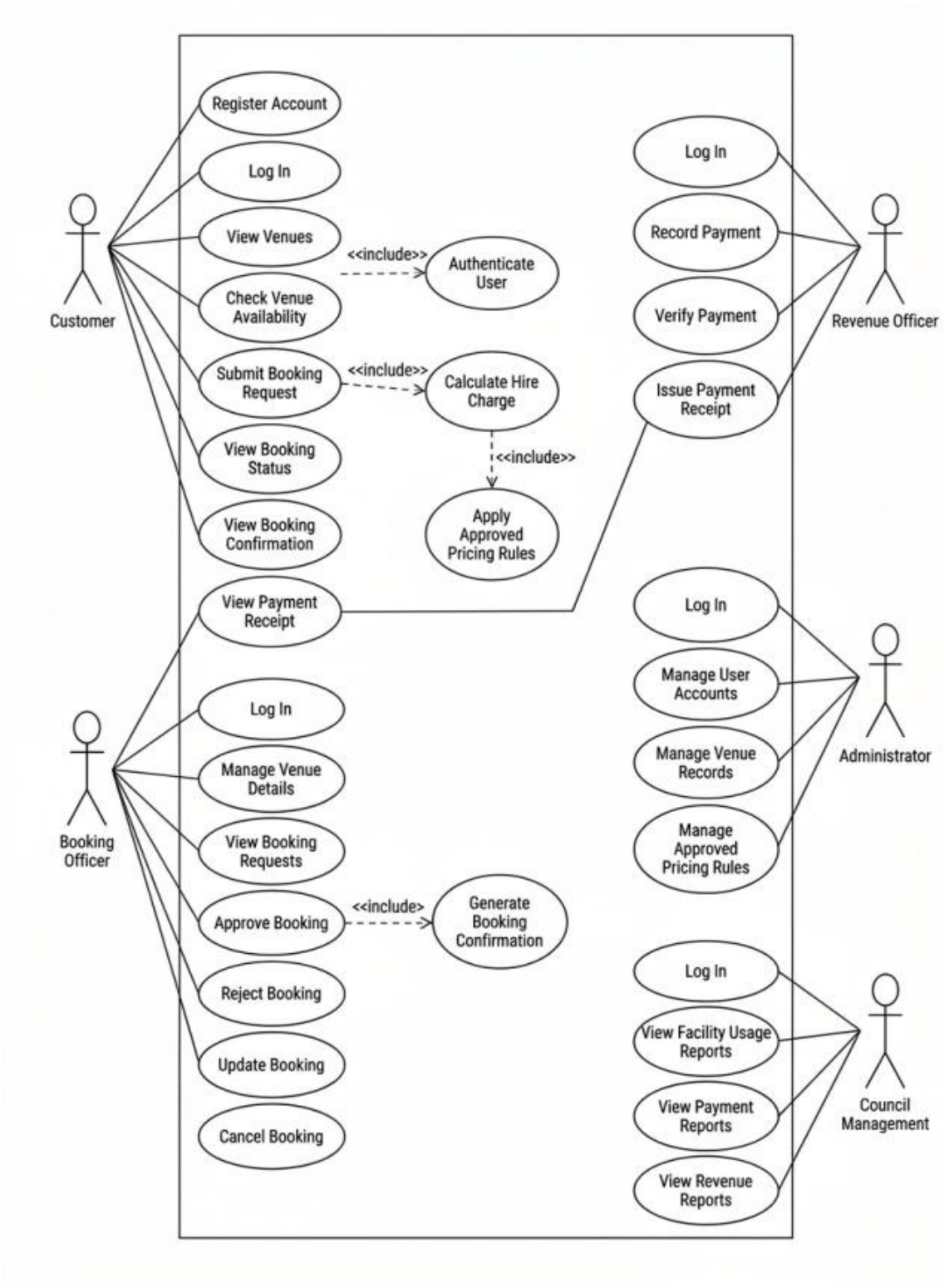


Figure 4.3.2 UML Use Case Diagram – COH-PMS

4.4 System Flowchart Diagram

The system flowchart illustrates the main processes of the City of Harare Property Management System (COH-PMS). It shows every system phase from customer login and venue selection to booking confirmation and receipt generation.

Figure 4.4.1 System Flowchart Diagram – COH-PMS

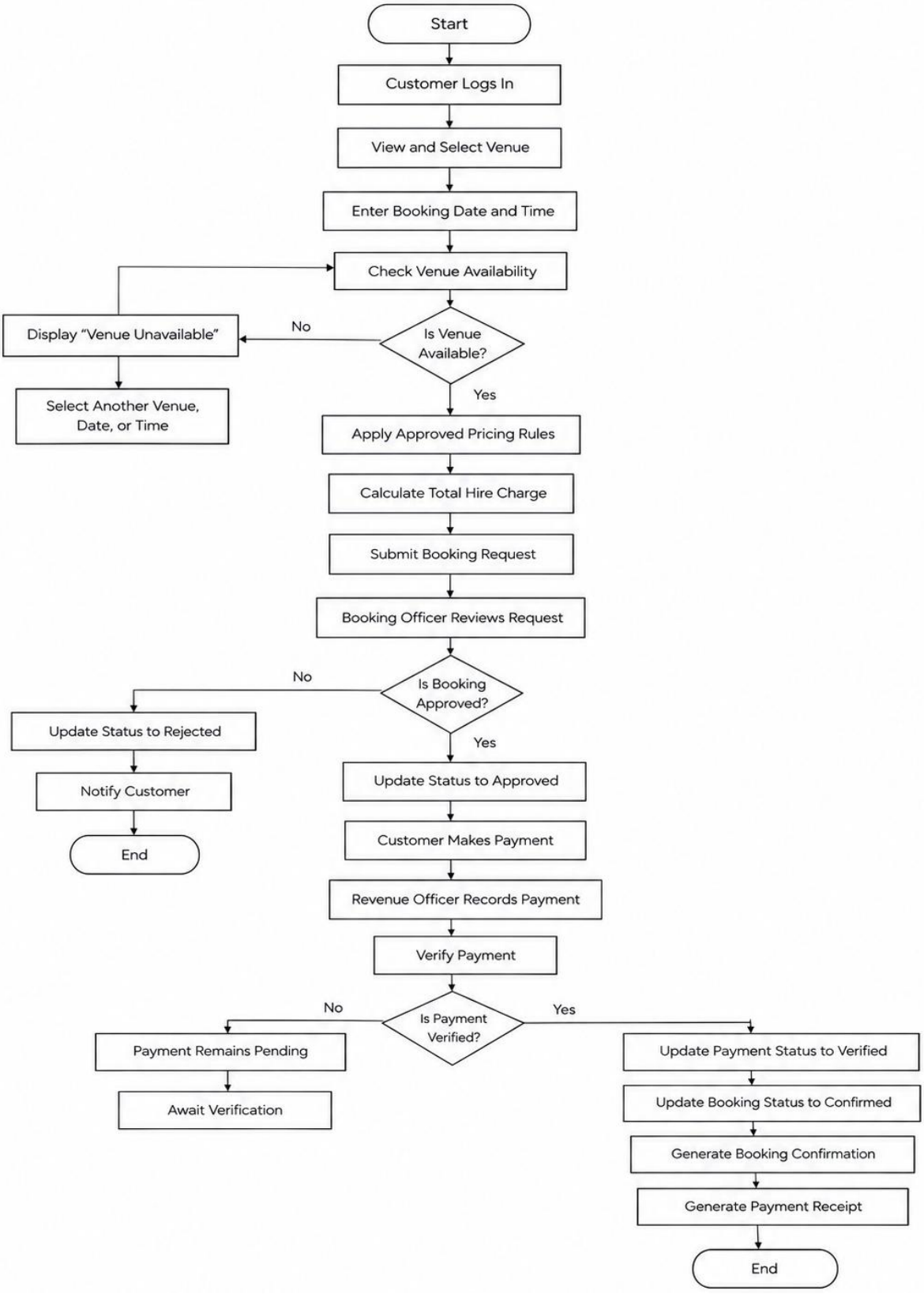


Figure 4.4.1 System Flowchart Diagram – COH-PMS

4.5 Sequence Diagram

The sequence diagram illustrates the main booking process in the City of Harare Property Management System (COH-PMS). It shows how the customer, booking officer, revenue officer, web application, and central database interact from venue selection to booking confirmation and receipt generation.

Figure 4.5.1 UML Sequence Diagram – COH-PMS

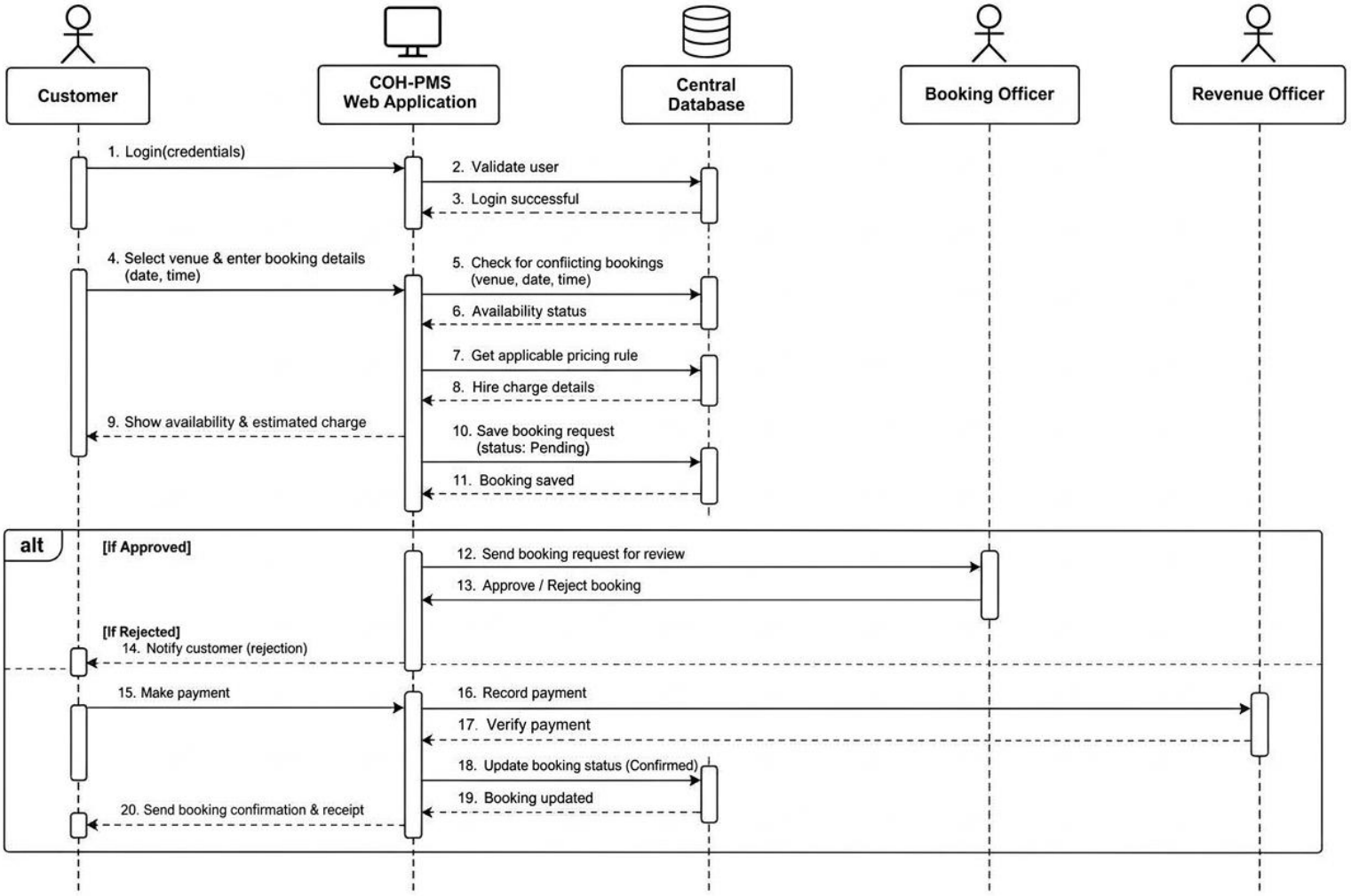


Figure 4.5.1 UML Sequence Diagram – COH-PMS

4.6 Database Design

The COH-PMS uses a central relational database to store and manage information about users, venues, bookings, approved pricing rules, and payments. The database design supports secure access, accurate venue information, prevention of double-booking, automated calculation of approved price adjustments, payment verification, and generation of facility-usage and revenue reports. The detailed data dictionary for the database records is presented in the following tables:

Table 4.6.1 Database Schema Summary

Record	Purpose	Main Relationships
Users	Stores customer and staff account details, login information, user roles, and account status.	One customer can create many bookings. Revenue officers can verify many payments.
Venues	Stores council property and venue details, including venue type, location, capacity, facilities, standard price, and status.	One venue can have many bookings on different dates and times.
Pricing Rules	Stores approved conditions for applying discounts or price increases to venue-hire charges.	One pricing rule can be applied to many bookings; a booking may have no pricing rule where no adjustment applies.
Bookings	Stores customer booking requests, selected venues, date and time details, event information, calculated charges, and booking status.	Each booking belongs to one customer and one venue. A booking may use one pricing rule and may have one payment.
Payments	Stores payment details, including booking reference, amount paid, payment method, transaction reference, verification information, and payment status.	Each payment is linked to one booking and is verified by one Revenue Officer.

The following class diagram shows the main classes of the City of Harare Property Management System (COH-PMS), their attributes, and the relationships between them.

Figure 4.6.2 UML Class Diagram – COH-PMS

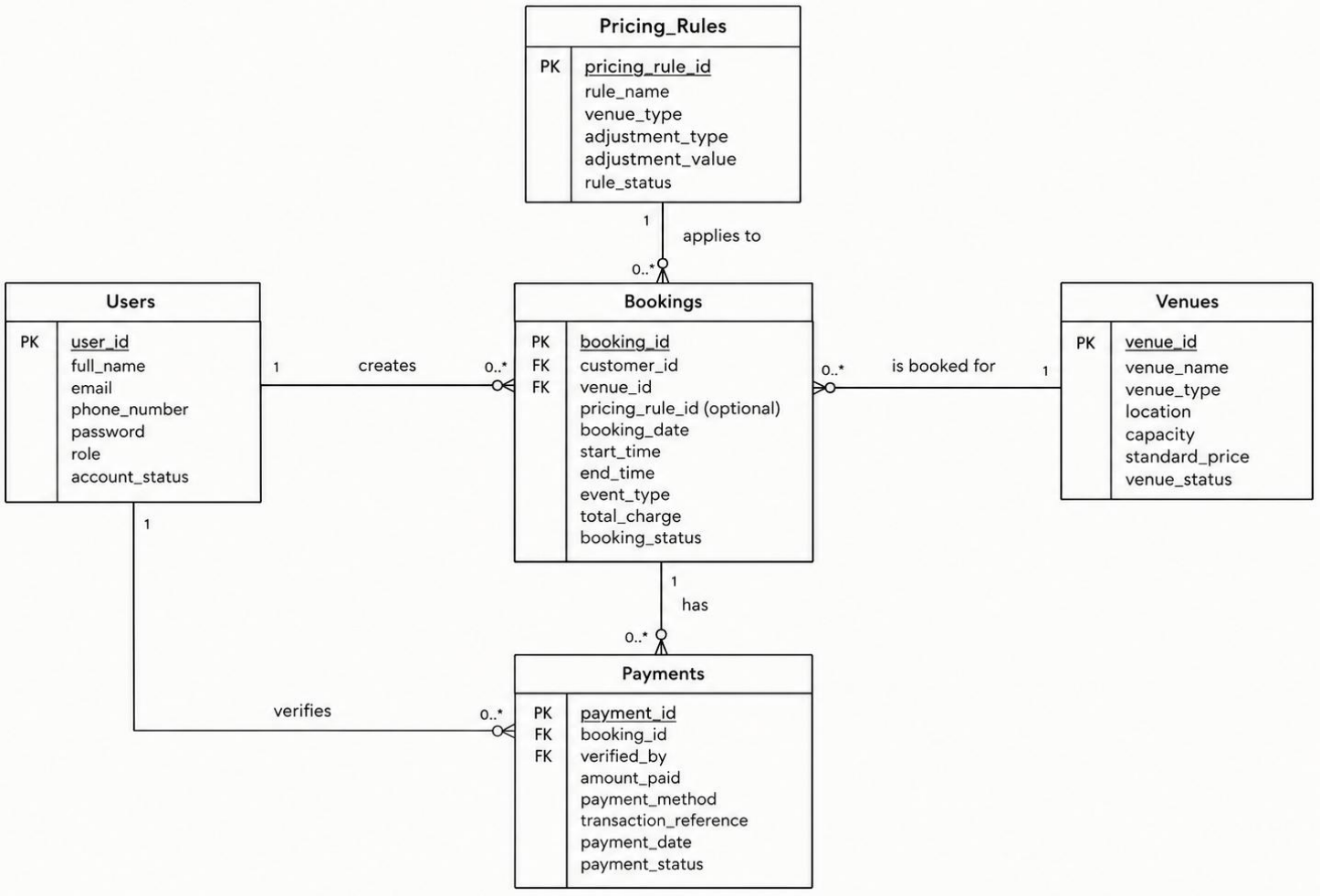


Figure 4.6.2 UML Class Diagram – COH-PMS

- The Entity Relationship Diagram shows the relationships between the Users, Venues, Pricing Rules, Bookings, and Payments records.
- One customer can create many bookings, while each booking belongs to one customer and one venue.
- A venue can be linked to many bookings on different dates and times.
- A booking may have one applied pricing rule, while one pricing rule can be applied to many bookings.
- Each payment is linked to one booking, and a revenue officer verifies payment records through the Users record.
- These relationships support availability checking, prevention of double-booking, approved pricing adjustments, payment verification, and the generation of facility-usage and revenue reports.

The detailed data dictionary for the database records is presented in the following tables:

Table 4.6.3 User Record

Attribute	Data Type	Key	Description
user_id	Integer	PK	Unique identifier for each system user
full_name	Varchar(100)		Full name of the customer or staff member
email	Varchar(100)		User email address used for login and communication
phone_number	Varchar(20)		User contact number
password	Varchar(255)		Encrypted user password
role	Varchar(30)		User role, such as Customer, Booking Officer, Revenue Officer, Administrator, or Council Management
account_status	Varchar(20)		Status of the account, such as Active or Inactive
created_at	DateTime		Date and time the account was created

Table 4.6.4 Venues Record

Attribute	Data Type	Key	Description
venue_id	Integer	PK	Unique identifier for each council property or venue
venue_name	Varchar(100)		Name of the venue or property
venue_type	Varchar(50)		Type of venue, such as hall, stadium, community centre, or open space
location	Varchar(100)		Physical location of the venue
capacity	Integer		Maximum number of people the venue can accommodate
facilities	Text		Available facilities or services at the venue
standard_price	Decimal(10,2)		Standard hire price before any approved adjustment
availability_status	Varchar(20)		Venue status, such as Available, Unavailable, or Under Maintenance
venue_status	Varchar(20)		Record status, such as Active or Inactive

Table 4.6.5 Pricing Rules Record

Attribute	Data Type	Key	Description
pricing_rule_id	Integer	PK	Unique identifier for each approved pricing rule
rule_name	Varchar(100)		Name of the approved pricing rule
venue_type	Varchar(50)		Type of venue to which the rule applies
condition_description	Text		Description of the condition for applying the rule
adjustment_type	Varchar(20)		Type of adjustment, such as Discount or Increase
adjustment_value	Decimal(10,2)		Percentage or fixed amount used for the price adjustment
rule_status	Varchar(20)		Status of the rule, such as Active or Inactive

Table 4.6.6 Bookings Record

Attribute	Data Type	Key	Description
booking_id	Integer	PK	Unique identifier for each booking request
customer_id	Integer	FK	References the customer in the Users record
venue_id	Integer	FK	References the selected venue in the Venues record
pricing_rule_id	Integer	FK	References the pricing rule applied to the booking; may be blank if no adjustment applies
booking_date	Date		Date on which the venue will be used
start_time	Time		Start time of the booking
end_time	Time		End time of the booking
event_type	Varchar(100)		Type or purpose of the event
expected_attendance	Integer		Expected number of people attending the event
standard_charge	Decimal(10,2)		Original venue-hire charge before adjustment
adjustment_amount	Decimal(10,2)		Amount added or deducted through the approved pricing rule
total_charge	Decimal(10,2)		Final booking amount payable
booking_status	Varchar(20)		Booking status: Pending, Approved, Rejected, Cancelled, or Confirmed
booking_created_at	DateTime		Date and time the booking request was submitted

Table 4.6.7 Payment Record

Attribute	Data Type	Key	Description
payment_id	Integer	PK	Unique identifier for each payment record
booking_id	Integer	FK	References the booking for which payment is made
amount_paid	Decimal(10,2)		Amount paid by the customer
payment_method	Varchar(50)		Method used to make payment, such as Cash, Bank Transfer, or Mobile Payment
transaction_reference	Varchar(100)		Payment or transaction reference number
payment_date	Date		Date on which payment was made or recorded
payment_status	Varchar(20)		Payment status, such as Pending, Verified, or Rejected
verified_by	Integer	FK	References the Revenue Officer in the Users record who verified the payment
verified_at	DateTime		Date and time the payment was verified

4.7 Pricing Rules Design

- The COH-PMS will automatically calculate property and venue hire charges using approved pricing rules. Each venue will have a standard hire price.
- When a customer selects a venue, booking date, and time, the system will check whether an active pricing rule applies and calculate the final hire charge.
- Only authorised administrators will create, update, activate, or deactivate pricing rules. Booking officers and customers will not be able to change approved pricing rules.
- The system will store the standard charge, adjustment amount, applied pricing rule, and final charge in the booking record to ensure that charges remain accurate and traceable.

The Final Charge will be calculated as:

Final Charge = Standard Price + Adjustment Amount

Figure 4.7.1 Flowchart – COH-PMS Pricing Rules Decision

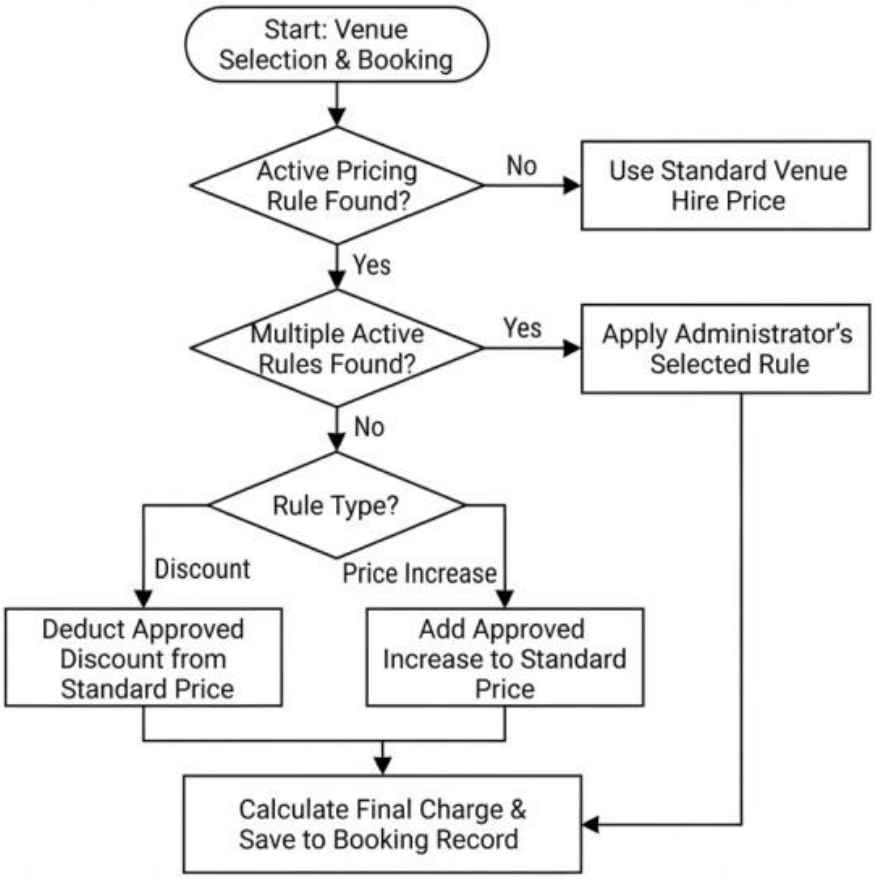


Figure 4.7.1 Flowchart – COH-PMS Pricing Rules Decision

4.8 Interface Design

The City of Harare Property Management System (COH-PMS) will use a simple web-based interface that can be accessed through a browser on a computer, tablet, or smartphone. The interface is user friendly, showing clear menus, labelled forms, validation messages, and role-based dashboards to ensure that customers and council staff can use the system easily.

Home Page

The home page will introduce the COH-PMS and provide links to view available venues, register an account, and log in. It will also provide brief guidance on how customers can check availability and submit booking requests.

Registration and Login Pages

The registration page will allow new customers to create accounts by entering their name, contact details, email address, and password. The login page will allow customers and authorised council staff to access their role-based dashboards using their email address and password.

Customer Venue Page

The customer venue page will display available council properties and venues. Each venue will show its name, type, location, capacity, available facilities, standard price, and availability status. Customers will be able to select a venue and proceed to check availability.

Booking Page

The booking page will allow customers to select a venue, booking date, start time, end time, event type, and expected attendance. The system will check availability, apply approved pricing rules, and display the calculated hire charge before the customer submits the booking request.

Booking Status Page

The customer booking-status page will display submitted booking requests and their statuses, including Pending, Approved, Rejected, Cancelled, or Confirmed. Customers will also be able to view booking confirmations and payment receipts where available.

Officer Dashboard

The booking officer dashboard will display venue records and booking requests. The booking officer will be able to add or update venue details, review booking requests, and approve, reject, update, or cancel bookings.

Officer Dashboard

The revenue officer dashboard will display approved bookings awaiting payment verification. The revenue officer will record payment details, verify payments, update payment status, and generate payment receipts.

Administrator Dashboard

The administrator dashboard will allow authorised administrators to manage user accounts, venue records, and approved pricing rules. Administrators will also be able to activate or deactivate user accounts, venues, and pricing rules where necessary.

Management Reports Dashboard

The management dashboard will provide reports on bookings, facility usage, payments, and revenue. Council management will be able to select report types and date ranges for viewing the required information.

4.9 Security Design

The COH-PMS will protect customer, booking, payment, pricing, and report information through user authentication, role-based access control, input validation, and regular database backups.

- User Authentication: Customers and council staff will log in using an email address and password. Passwords will be stored securely in encrypted form.
- Role-Based Access Control: Users will only access functions relevant to their roles. Customers will manage their own booking requests, booking officers will manage bookings, revenue officers will record and verify payments, and administrators will manage users, venues, and pricing rules, while council management will access reports.
- Input Validation: The system will validate required fields, dates, times, amounts, and booking details before saving information. It will prevent invalid bookings and reduce incorrect data entry.
- Booking Protection: The system will check venue availability before allowing a booking request to be submitted. This will reduce the risk of double-booking for the same venue, date, and time.
- Pricing Rule Protection: Only authorised administrators will create, update, activate, or deactivate approved pricing rules. The system will record the pricing rule and final charge applied to each booking.
- Payment Protection: Only revenue officers will record and verify payment details. Each payment will be linked to a valid booking and include a transaction reference.
- Data Backup and Recovery: The central database will be backed up regularly to reduce the risk of data loss caused by system failure, accidental deletion, or hardware problems.
- Session Control: The system will allow users to log out after completing their tasks, reducing the risk of unauthorised access on shared devices.

4.10 Conclusion

This chapter presented the system design for the City of Harare Property Management System (COH-PMS). The design defined the three-layer system architecture, user interactions, booking and payment flow, database records, class relationships, sequence of events, pricing rules, interface design, and security measures. The design supports the management of council properties and venues, prevents double-booking through availability checking, applies approved pricing rules when calculating hire charges, records and verifies payments, and generates reports on facility usage and revenue. The next chapter presents the implementation of the COH-PMS, including the technologies and development tools used to build the system.

CHAPTER V: CODING AND TESTING

5.0 Introduction

This chapter presents the coding and testing of the City of Harare Property Management System (COH-PMS). It describes how the system design presented in Chapter 4 was translated into a working web-based application using the selected programming tools and database technology. The chapter also presents the testing procedures used to verify that the system performs its required functions correctly before implementation.

5.1 Programming Languages and Technologies Used

The COH-PMS was developed as a web-based application using HTML, CSS, JavaScript, PHP, and MySQL. These technologies were selected because they support the development of a secure, interactive, and database-driven system that can be accessed through a web browser. The development and testing of the system were carried out using Laragon and Visual Studio Code.

- HTML: For creating the structure of the COH-PMS web pages by developing the login page, registration page, venue management pages, booking pages, payment pages, and administrative report pages.
- CSS: Applied styling to control the layout, colours, fonts, buttons, tables, navigation menus, and overall appearance of the system. CSS also helped to make the system pages clear, organised, and suitable for use on different screen sizes.
- JavaScript: To validate user input, display confirmation messages, control date and time selection, and support venue availability checking before a booking is submitted.
- PHP: Server-side programming language for the COH-PMS, to process booking requests, connect the system to the MySQL database, manage user login and access control, calculate booking charges, apply approved pricing rules, record payments, and generate facility-usage and revenue reports.
- MySQL: For database management system for the COH-PMS, to create and manage the Users, Venues, Pricing Rules, Bookings, and Payments tables. MySQL stored customer details, venue information, booking records, pricing rules, and payment records, and supported the relationships presented in the Entity Relationship Diagram.
- Visual Studio Code: The code IDE for developing the COH-PMS, to write, edit, and organise the HTML, CSS, JavaScript, PHP, and SQL files used in the system. Visual Studio Code also supported efficient coding through features such as syntax highlighting, file management, and error identification.
- Laragon: Provided Apache as the web server, PHP for server-side processing, and MySQL for database management. Laragon allowed the system to be developed and tested on a local computer before it was deployed for use by authorised City of Harare staff and customers.

5.2 Application File Structure

The COH-PMS was organised using a MVC-style source code structure to separate the system's configuration, business logic, database access, user interface, and public files. This structure improves code organisation, makes maintenance easier, and ensures that each part of the system performs a specific function.

Figure 5.2.1 Summary of Application File Structure

Folder / File	Description
app/controllers/	Handles user requests for authentication, users, venues, pricing rules, bookings, payments, reports, and dashboards.
app/core/	Contains the router and base controller, including authentication and role-based access checks.
app/helpers/	Stores reusable functions for authentication, CSRF protection, flash messages, formatting, validation, and file uploads.
app/models/	Contains database-access classes that use prepared PDO statements for users, venues, pricing rules, bookings, payments, and reports.
app/services/	Contains business logic for bookings, availability checking, pricing rules, payment verification, and reports.
config/	Stores application settings, role and status constants, and the secure PDO database connection.
database/	Contains the SQL files used to create the COH-PMS database and insert sample data.
docs/	Stores the system specification, testing records, screenshot list, and documentation of system changes.
public/	The only folder exposed to the web server; contains the front controller, assets, and uploaded venue images.
public/assets/	Contains CSS, JavaScript, and image files used for the user interface and interactive booking features.
public/uploads/venues/	Stores uploaded venue photographs, with restrictions to prevent script execution.
tests/	Contains test files for pricing logic, security controls, validation, and business-rule transitions.
views/	Contains user-interface pages for authentication, customers, staff, errors, layouts, and the home page.
.htaccess	Restricts direct web access to sensitive folders such as app, config, database, docs, tests, and views.
README.md	Provides setup instructions, login details, and an overview of the system architecture.

5.3 Application Key Source Codes

- Secure Venue Management:
  Venue management was secured on the server to ensure that only authorised City of Harare staff could create or update venue records. The VenueController restricted venue creation and editing to the Administrator and Booking Officer roles. It also accepted only POST requests for changes, verified the CSRF token, and validated venue details before saving them to the database.

  Role check, POST-only restriction, CSRF verification, and input validation for venue creation:

```
public function create(): void
{
    if ($this->denyIfNotRole([ROLE_ADMINISTRATOR, ROLE_BOOKING_OFFICER])) {
        return;
    }

    if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
        $this->redirect(BASE_URL . '/?r=venue/index');
    }

    if (!verify_csrf_token($_POST['csrf_token'] ?? null)) {
        error_log('CSRF validation failed for venue/create.');
        set_flash('error', 'Your session has expired.');
        $this->redirect(BASE_URL . '/?r=venue/index');
    }

    $validation = $this->validateVenue($_POST);
    if ($validation['errors'] !== []) {
        set_flash('error', implode(' ', $validation['errors']));
        $this->redirect(BASE_URL . '/?r=venue/index');
    }

    $data = $validation['data'];
    $savedImage = null;
    if (!empty($_FILES['image']['name'])) {
        $savedImage = save_uploaded_venue_image($_FILES['image']);
        if ($savedImage === null) {
            set_flash('error', 'Image upload failed. Use JPG, PNG, or WEBP under 2MB.');
            $this->redirect(BASE_URL . '/?r=venue/index');
        }
    }

    $data['image_path'] = $savedImage;
    try {
        Venue::create($data);
    } catch (Throwable $exception) {
        $this->deleteVenueImage($savedImage);
        error_log('Venue creation failed: ' . $exception->getMessage());
        set_flash('error', 'Venue could not be saved. Please try again.');
        $this->redirect(BASE_URL . '/?r=venue/index');
    }
```

- Automatic Pricing Rule:
  The system automatically calculated the final booking charge without manual calculation. It selected the applicable active pricing rule based on the venue type, booking date, day of the week, and booking duration, then applied the approved discount or surcharge to the venue's standard price. This implementation delivers the second project objective of automatically adjusting property prices using approved rules.

  Pricing Rule Selection and Charge Calculation:

```
public function calculate(int $venueId, string $date, string $start, string $end): array

  {
    $venue = ($this->venueFinder)($venueId);
    if (!$venue) {
      throw new InvalidArgumentException('Venue not found.');
    }

    $startMinutes = $this->timeToMinutes($start);
    $endMinutes = $this->timeToMinutes($end);
    if ($endMinutes <= $startMinutes) {
      throw new InvalidArgumentException('End time must be later than start time.');
    }

    $durationHours = ($endMinutes - $startMinutes) / 60;
    $standardCharge = round((float) $venue['standard_price'], 2);
    $selectedRule = null;

    foreach (($this->activeRulesFinder)() as $rule) {
      if (($rule['rule_status'] ?? 'Active') !== 'Active') {
        continue;
      }
      if ($rule['venue_type'] !== 'All' && $rule['venue_type'] !== $venue['venue_type']) {
        continue;
      }
      if (!$this->ruleMatchesDate($rule, $date)) {
        continue;
      }
      if (!empty($rule['days_of_week']) && !$this->ruleMatchesDay($rule['days_of_week'], $date)) {
        continue;
      }
      if (!empty($rule['min_hours']) && $durationHours < (float) $rule['min_hours']) {
        continue;
      }

      $priority = (int) $rule['priority'];
      if (
        $selectedRule === null
        || $priority > (int) $selectedRule['priority']
        || ($priority === (int) $selectedRule['priority']
          && (int) $rule['pricing_rule_id'] < (int) $selectedRule['pricing_rule_id'])
      ) {
        $selectedRule = $rule;
      }
    }
```

```
        $adjustmentAmount = $selectedRule === null
            ? 0.0
            : $this->applyAdjustment($standardCharge, $selectedRule);
        $totalCharge = max(0.0, $standardCharge + $adjustmentAmount);

        return [
            'standard_charge' => round($standardCharge, 2),
            'adjustment_amount' => round($adjustmentAmount, 2),
            'total_charge' => round($totalCharge, 2),
            'pricing_rule_id' => $selectedRule['pricing_rule_id'] ?? null,
            'rule_name' => $selectedRule['rule_name'] ?? null,
        ];
    }
```

Discount and Surcharge Calculation:

```
 private function applyAdjustment(float $standardCharge, array $rule): float

    {
        $directions = [
            'Weekday Discount' => 'discount',
            'Off-Peak Discount' => 'discount',
            'Long Booking Discount' => 'discount',
            'Weekend Surcharge' => 'surcharge',
            'Peak Demand Surcharge' => 'surcharge',
        ];
        $direction = $directions[$rule['rule_type']] ?? null;
        if ($direction === null) {
            throw new InvalidArgumentException('Pricing rule type is invalid.');
        }

        $value = (float) $rule['adjustment_value'];
        $amount = $rule['adjustment_type'] === 'Fixed Amount'
            ? $value
            : $standardCharge * ($value / 100);

        if ($direction === 'discount') {
            return -min($standardCharge, $amount);
        }

        return $amount;
    }
}
```

- Facility Usage and Revenue Reporting:
  The system generated facility-usage and revenue reports from stored booking and payment records. Facility usage was calculated from approved, confirmed, and completed bookings, while revenue was calculated only from payments verified by a Revenue Officer. This implementation delivers the third project objective of providing reports on facility usage and revenue.

  Facility Usage Calculation:

```
public function getUsageReport(array $filters = []): array

  {
    $rows = Report::getUsageReport($filters);
    $bounds = Report::getUsageDateBounds($filters);
    $startDate = !empty($filters['start_date']) ? (string) $filters['start_date'] : ($bounds['start_date'] ?? null);
    $endDate = !empty($filters['end_date']) ? (string) $filters['end_date'] : ($bounds['end_date'] ?? null);
    $days = $this->getInclusiveDays($startDate, $endDate);
    $availableHoursPerVenue = $days * self::AVAILABLE_HOURS_PER_DAY;

    $bookingCount = 0;
    $totalHours = 0.0;
    $availableHours = 0.0;
    foreach ($rows as &$row) {
      $row['booking_count'] = (int) $row['booking_count'];
      $row['total_hours'] = round((float) $row['total_hours'], 2);
      $row['available_hours'] = $availableHoursPerVenue;
      $row['usage_percentage'] = $availableHoursPerVenue > 0
        ? round(($row['total_hours'] / $availableHoursPerVenue) * 100, 2)
        : 0.0;
      $bookingCount += $row['booking_count'];
      $totalHours += $row['total_hours'];
      $availableHours += $availableHoursPerVenue;
    }
    unset($row);

    return [
      'rows' => $rows,
      'totals' => [
        'booking_count' => $bookingCount,
        'total_hours' => round($totalHours, 2),
        'available_hours' => round($availableHours, 2),
        'usage_percentage' => $availableHours > 0
          ? round(($totalHours / $availableHours) * 100, 2)
          : 0.0,
      ],
      'range_days' => $days,
    ];
  }
```

Verified Revenue Report:

```
public function getRevenueReport(array $filters = []): array

  {
     $rows = Report::getRevenueReport($filters);
     $monthlyRows = Report::getRevenueByMonth($filters);
     $totalAmount = array_sum(array_map(static fn(array $row): float => (float) $row['total_amount'],
$rows));
     $maxAmount = max([0.0, ...array_map(static fn(array $row): float => (float) $row['total_amount'],
$rows)]);

     foreach ($rows as &$row) {
        $amount = (float) $row['total_amount'];
        $row['total_amount'] = round($amount, 2);
        $row['payment_count'] = (int) $row['payment_count'];
        $row['bar_percentage'] = $maxAmount > 0 ? (int) round(($amount / $maxAmount) * 100) : 0;
     }
     unset($row);

     foreach ($monthlyRows as &$row) {
        $row['payment_count'] = (int) $row['payment_count'];
        $row['total_amount'] = round((float) $row['total_amount'], 2);
     }
     unset($row);

     return [
        'rows' => $rows,
        'by_month' => $monthlyRows,
        'totals' => [
           'payment_count' => array_sum(array_column($rows, 'payment_count')),
           'total_amount' => round($totalAmount, 2),
        ],
     ];
  }
```

- Payment Verification
  The system allowed only an Administrator or Revenue Officer to verify a payment. Verification was permitted only when the payment status was Pending Verification. During verification, the system locked the payment and its related booking within a database transaction, confirmed that the booking status was approved, generated a unique receipt number, changed the payment status to Verified, and updated the booking status to Confirmed. If any error occurred, the transaction was rolled back to maintain data consistency.

```
public function verify(int $paymentId, int $actorId, string $actorRole): void
  {
     $this->setDecision($paymentId, PAYMENT_STATUS_VERIFIED, $actorId, $actorRole);
  }

  public function reject(int $paymentId, int $actorId, string $actorRole): void
  {
     $this->setDecision($paymentId, PAYMENT_STATUS_REJECTED, $actorId, $actorRole);
  }

  private function setDecision(int $paymentId, string $status, int $actorId, string $actorRole): void
  {
     if (!in_array($actorRole, [ROLE_ADMINISTRATOR, ROLE_REVENUE_OFFICER], true)) {
        throw new DomainException('You are not allowed to verify or reject payments.');
     }

     $pdo = get_db_connection();
     $pdo->beginTransaction();
     try {
        $payment = Payment::findByIdForUpdate($paymentId, $pdo);
        if (!$payment) {
           throw new DomainException('Payment record not found.');
        }
        if ($payment['payment_status'] !== PAYMENT_STATUS_PENDING_VERIFICATION) {
           throw new DomainException('Only a payment pending verification can be updated.');
        }

        $booking = Booking::findByIdForUpdate((int) $payment['booking_id'], $pdo);
        if (!$booking) {
           throw new DomainException('The payment booking could not be found.');
        }

        if ($status === PAYMENT_STATUS_VERIFIED) {
           if ($booking['booking_status'] !== BOOKING_STATUS_APPROVED) {
              throw new DomainException('Only an approved booking can be confirmed by payment
verification.');
           }
           $this->verifyPaymentWithReceipt($paymentId, $actorId, $pdo);
           (new BookingService())->changeStatus(
              (int) $booking['booking_id'],
              BOOKING_STATUS_CONFIRMED,
              $actorId,
              $actorRole,
              true,
              $pdo
           );
        } else {
```

```
            if (!Payment::updateStatus($paymentId, PAYMENT_STATUS_REJECTED, $actorId, null, $pdo)) {
                throw new RuntimeException('Payment rejection could not be saved.');
            }
        }

        $pdo->commit();
    } catch (Throwable $exception) {
        if ($pdo->inTransaction()) {
            $pdo->rollBack();
        }
        throw $exception;
    }
  }

  private function verifyPaymentWithReceipt(int $paymentId, int $actorId, PDO $pdo): void
  {
    for ($attempt = 0; $attempt < 5; $attempt++) {
        $receiptNumber = $this->nextReceiptNumber($pdo);
        try {
            if (!Payment::updateStatus($paymentId, PAYMENT_STATUS_VERIFIED, $actorId, $receiptNumber,
$pdo)) {
                throw new RuntimeException('Payment verification could not be saved.');
            }
            return;
        } catch (PDOException $exception) {
            if ((int) ($exception->errorInfo[1] ?? 0) !== 1062 || $attempt === 4) {
                throw $exception;
            }
        }
    }

    throw new RuntimeException('A unique receipt number could not be generated.');
  }

  private function nextReceiptNumber(PDO $pdo): string
  {
    $prefix = 'RCT-' . date('Ymd') . '-';
    $stmt = $pdo->prepare('SELECT receipt_number FROM payments WHERE receipt_number LIKE
:receipt_prefix ORDER BY receipt_number DESC LIMIT 1 FOR UPDATE');
    $stmt->execute([':receipt_prefix' => $prefix . '%']);
    $lastReceipt = $stmt->fetchColumn();
    $sequence = $lastReceipt === false ? 1 : ((int) substr((string) $lastReceipt, -4)) + 1;

    return $prefix . str_pad((string) $sequence, 4, '0', STR_PAD_LEFT);
  }
}
```

- Booking Reference Generation
  The system generated a unique reference for each new booking after validating the venue, booking date, time, capacity, availability, and calculated charge. The reference followed the format COH-YYYYMMDD-XXXX, where the date section represented the booking date and the four-digit sequence identified the booking number for that date. The system first saved the booking with a temporary reference, then replaced it with the final unique reference within the same database transaction. If a duplicate reference occurred, the system retried up to five times before returning an error.

```
private function assignBookingReference(int $bookingId, string $bookingDate, PDO $pdo): void

  {
    $datePart = str_replace('-', '', $bookingDate);
    for ($attempt = 0; $attempt < 5; $attempt++) {
      $highestSequence = 0;
      foreach (Booking::getReferencesForDate($bookingDate, $bookingId, $pdo) as $reference) {
        if (preg_match('/^COH-' . preg_quote($datePart, '/') . '-(\d{4})$/', (string) $reference, $matches)) {
          $highestSequence = max($highestSequence, (int) $matches[1]);
        }
      }

      $reference = 'COH-' . $datePart . '-' . str_pad((string) ($highestSequence + 1), 4, '0', STR_PAD_LEFT);
      try {
        if (!Booking::updateReference($bookingId, $reference, $pdo)) {
          throw new RuntimeException('Booking reference could not be saved.');
        }
        return;
      } catch (PDOException $exception) {
        if ((int) ($exception->errorInfo[1] ?? 0) !== 1062 || $attempt === 4) {
          throw $exception;
        }
      }
    }

    throw new RuntimeException('A unique booking reference could not be generated.');
  }
}
```

5.4 Testing

This section presents the testing procedures applied during and after the development of the COH-PMS. Testing was conducted to verify that each module functions correctly in isolation and that the integrated system meets the functional and non-functional requirements. The tests covered include unit testing, module testing, system testing, and security testing:

Unit Testing

- Unit testing was conducted on individual functions of the COH-PMS using the live system database, without completing the full booking workflow.
- The Administrator tested the automated pricing calculation using the approved municipal pricing rules for Mbare Community Hall, which had a standard rate of US$ 100.00.
- The test confirmed that the system correctly applied a 10% weekday discount, producing a total charge of US$ 90.00, a 20% weekend surcharge, producing US$ 120.00, and a 5% long-booking discount, producing US$ 95.00.
- The booking-reference format was also tested to confirm that generated references followed the required COH-YYYYMMDD-XXXX format.
- All unit tests passed.

Module Testing

- Module testing was conducted to verify that related functions worked together correctly. Customer selected Mbare Community Hall for 20 October 2026 from 09:00 to 13:00.
- The system checked that the time slot was available, calculated the weekday discounted charge of US$ 90.00, created the booking, and generated the reference COH-20261020-0002.
- A second availability check for the same time slot was rejected because it overlapped with the existing booking.
- The system also rejected a booking request for 500 attendees because the hall capacity was 180.
- In addition, the Customer attempted to submit payment for a booking still marked Pending, and the system correctly rejected the payment because payments can only be submitted after a Booking Officer approves the booking, all module tests passed.

System Testing

- System testing was conducted by completing the full booking-to-payment-to-report workflow using the actual system user roles.
- Customer created a booking for Kuwadzana Community Centre on 6 November 2026 from 10:00 to 14:00 for 100 attendees.
- The system applied the relevant pricing rule and created booking reference COH-20261106-0001 with a total charge of US$ 144.00 and a Pending status.
- Booking Officer reviewed and approved the booking.
- The Customer then submitted a Bank Transfer payment using transaction reference CBZ-TRX-2026-9921, which was saved as Pending Verification.
- Revenue Officer verified the payment, generated receipt number RCT-20261005-0001, and changed the booking status from Approved to Confirmed.
- Finally, Council Management user, Council Secretary viewed the reports and confirmed that verified revenue increased from US$ 669.00 to US$ 813.00 and that the venue recorded four booked hours, representing 33.33% usage of a 12-hour available day.
- All system tests passed, confirming that the system correctly enforced role permissions, maintained transaction records, and updated reports.

Acceptance Testing

- Acceptance testing was conducted using the actual system user roles to confirm that the COH-PMS met its main requirements.
- The Customer successfully created a booking and submitted payment details, Booking Officer approved the booking, and Revenue Officer verified the payment and confirmed the booking, while Council Management user, Council Secretary viewed the booking, revenue, and facility-usage reports.
- The Administrator successfully managed venues, user accounts, and pricing rules.
- The test confirmed that each user could perform their authorised tasks and that the system was suitable for operational use.

Security Testing

- Security testing was conducted to confirm that users could only access functions assigned to their roles.
- The Customer could manage only their own bookings and could not approve bookings or verify payments. Booking Officer could approve bookings but could not verify payments, while Revenue Officer could verify payments but could not create bookings.
- Council Management had read-only access to reports.
- The system also prevented payment verification for a booking that was not approved and rejected duplicate or invalid payment actions.
- The tests confirmed that the system enforced role-based access control and protected booking and payment records.

5.5 Conclusion

This chapter presented the coding and testing procedures applied during the development of the COH-PMS. The system passed all unit, module, system, acceptance, and security tests, confirming that its key functions, user access controls, and data-processing procedures operated correctly. The next chapter presents the implementation of the system, including deployment, user training, and the changeover from the manual booking process to the COH-PMS.

CHAPTER VI: IMPLEMENTATION

6.0 Introduction

This chapter presents the implementation of the tested City of Harare Property Management System (COH-PMS). Implementation is the process of deploying the completed system so that authorised users can begin using it in their daily work. The chapter explains the deployment of the application and database, presents the main interfaces for different user roles, and outlines user training, system changeover, and on-going technical support.

6.1 System Deployment

- Deployment Environment:
  The COH-PMS is deployed as a web-based application using a PHP-enabled server environment. The system uses PHP for server-side processing, MySQL for data storage, and HTML, CSS, and JavaScript for the user interface. The application is accessed through a modern web browser by authorised customers and Council staff. During development and initial deployment, the system in hosted in the Laragon local server environment. Laragon provided the web server, PHP runtime, and MySQL database services required to run the application. This environment enables the system to be tested and prepared for full deployment to a Council-hosted or public web server.

- Application And Database deployment:
  The application files are placed in the web server directory and configured to connect to the coh_pms MySQL database. The database structure is deployed by creating the required tables, relationships, indexes, and constraints for users, venues, bookings, payments, pricing rules, and reports. Initial system data is then loaded into the database. This includes authorised user accounts, venue details, venue capacities, standard charges, approved pricing rules, and user-role information. The database connection settings are stored in the system configuration file to enable the application to communicate securely with the MySQL database. After deployment, the system is opened through the browser to confirm that the login page, database connection, navigation menu, and main user functions were working correctly.

- User Account Setup and Role Assignment:
  User accounts are created and assigned according to the responsibilities of each system user. Each account is given a role that control the functions and information available after login:

  Table 6.1.1 Role, Deployment Responsibility and Access

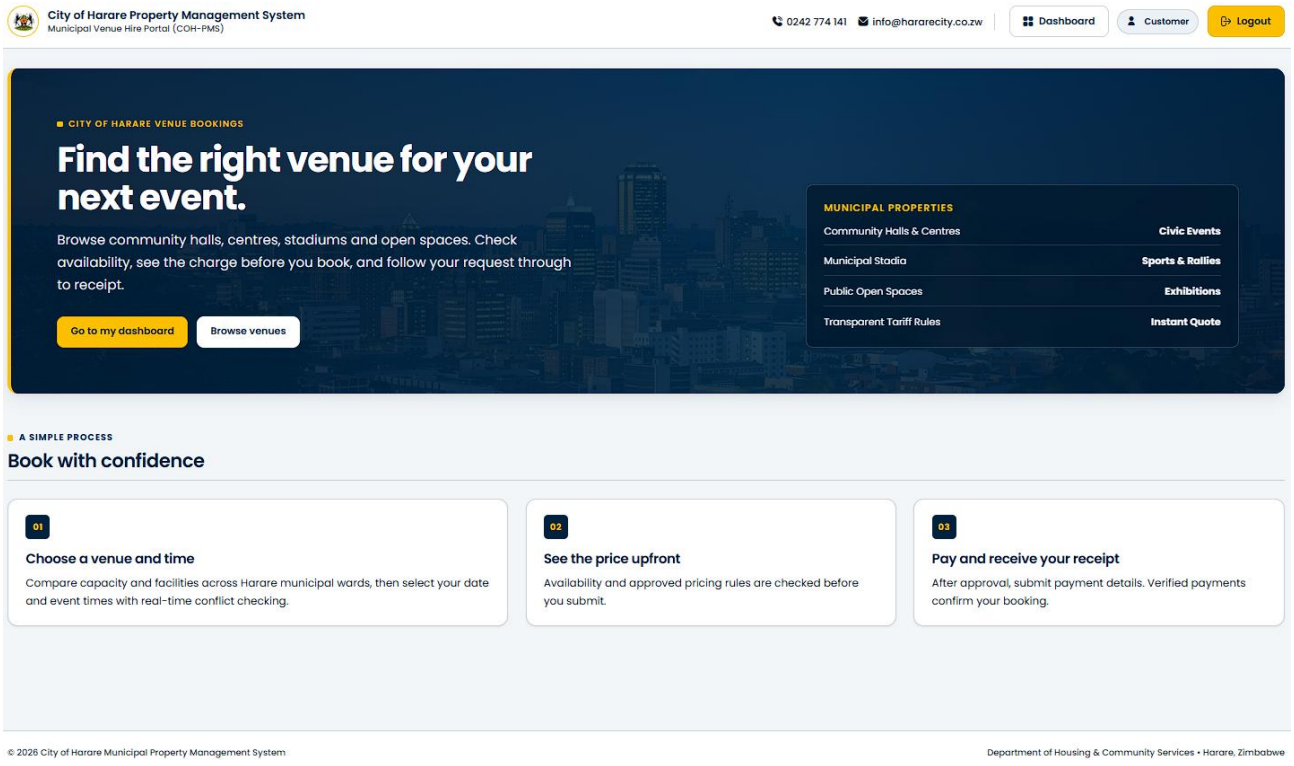
User role	Deployment responsibility and system access
Customer	Registers an account, views available venues, submits bookings, tracks booking status, submits payment details, and cancels eligible bookings.
Booking Officer	Reviews booking requests, checks booking information, and approves, rejects, or cancels bookings according to Council procedures.
Revenue Officer	Reviews payments awaiting verification, verifies or rejects payments, and issues receipt numbers for verified payments.
Council Management	Accesses read-only dashboard information, booking summaries, facility-usage reports, payment reports, revenue reports, and CSV exports.
Administrator	Manages user accounts, venues, approved pricing rules, and administrative system information.

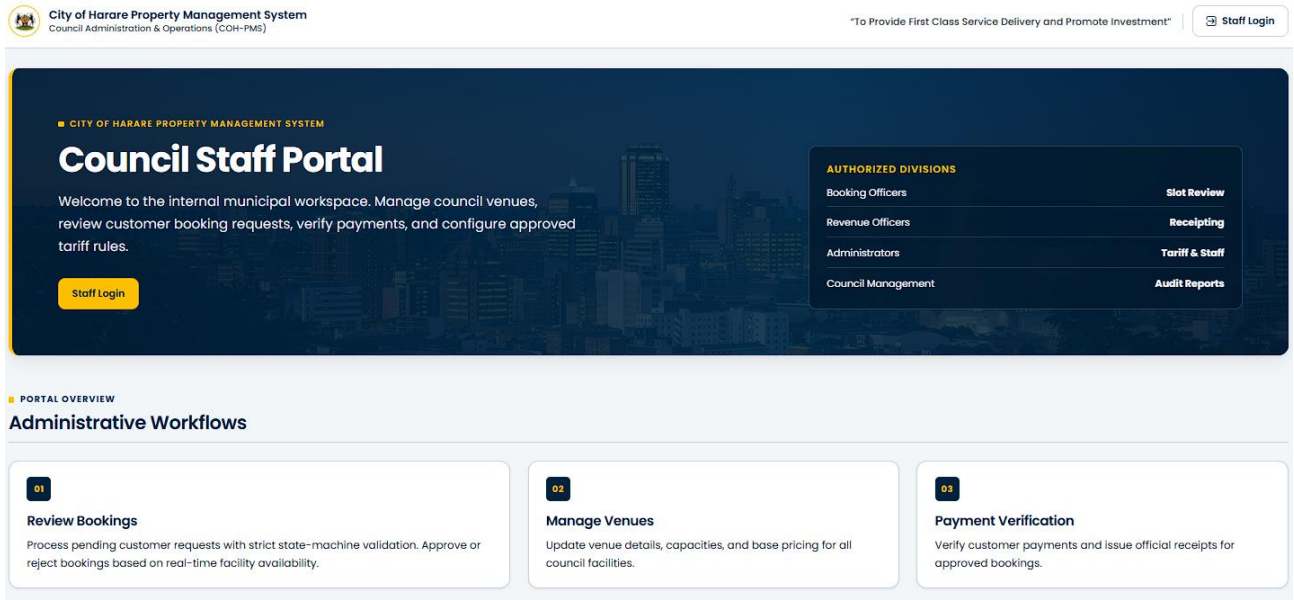
6.2 Operational System Interface Demonstration

The operational system interface demonstration presents the main interfaces available after deployment of the COH-PMS.

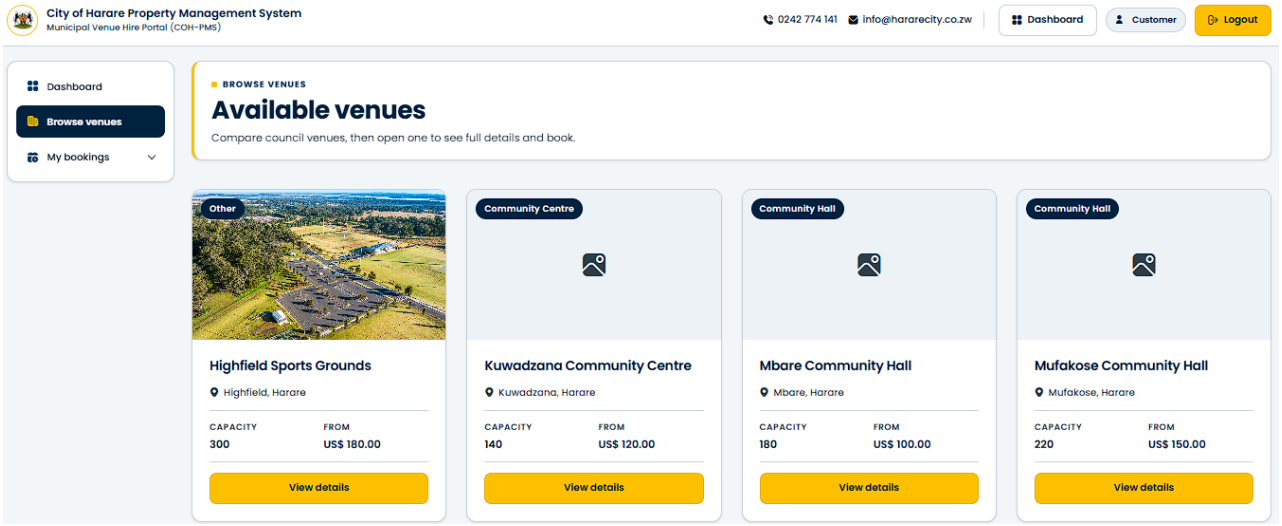
Landing Page:

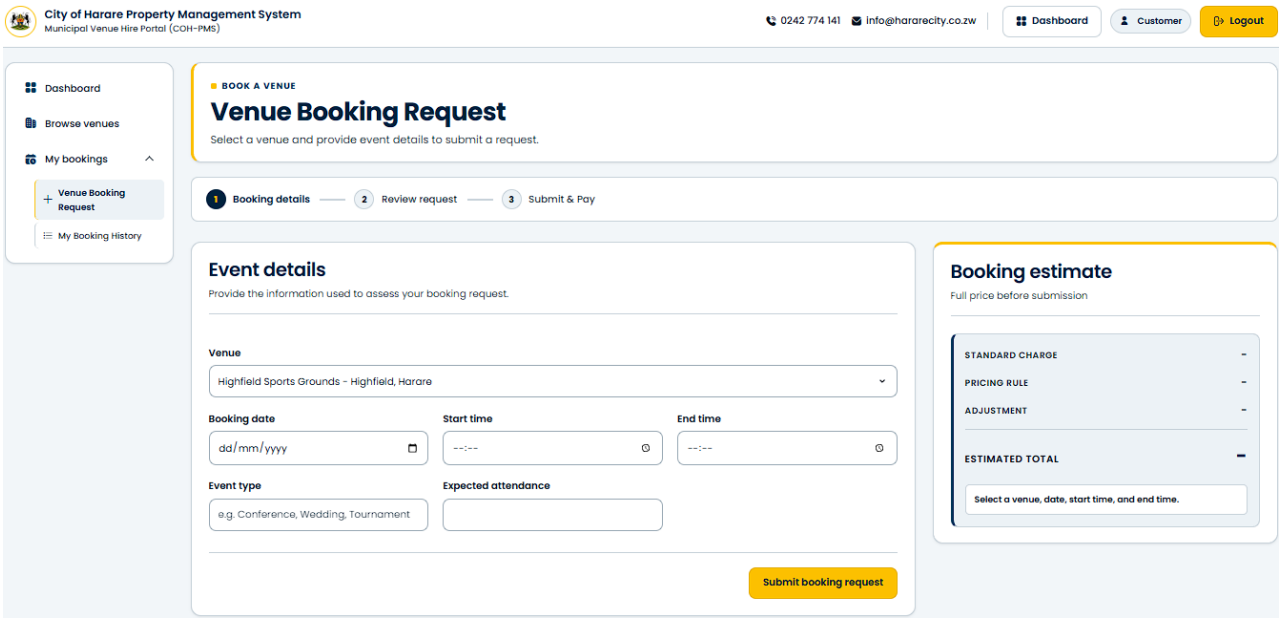
The landing page is the first page displayed when users visit the COH-PMS through a web browser. It introduces the system, presented the main services offered by the City of Harare Property and Venue Hire Management System, and provides navigation links for customers and council staff.

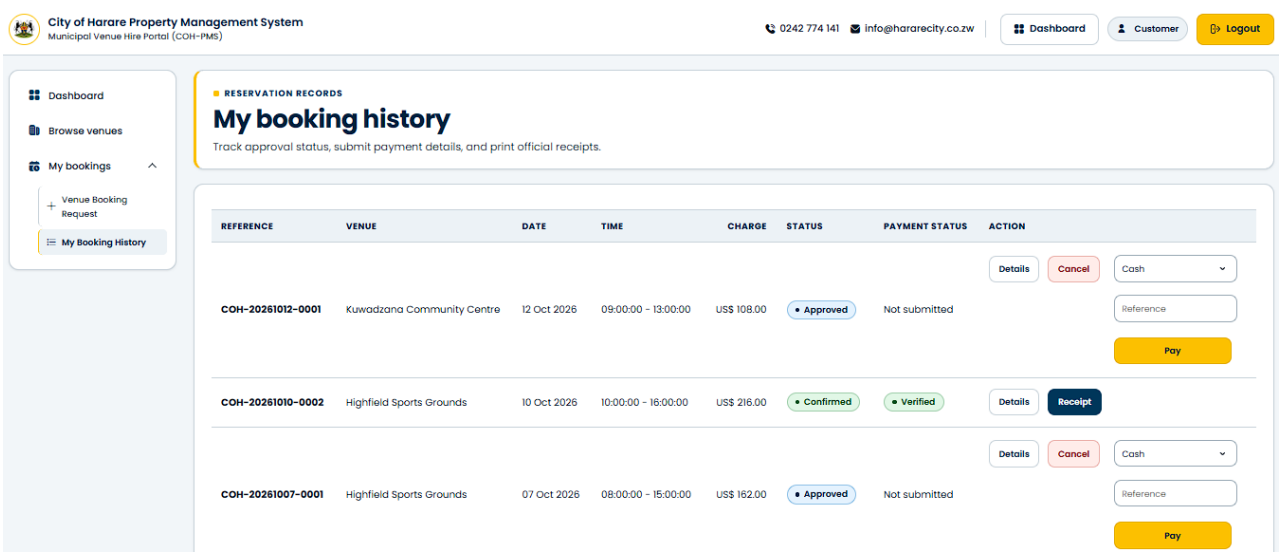




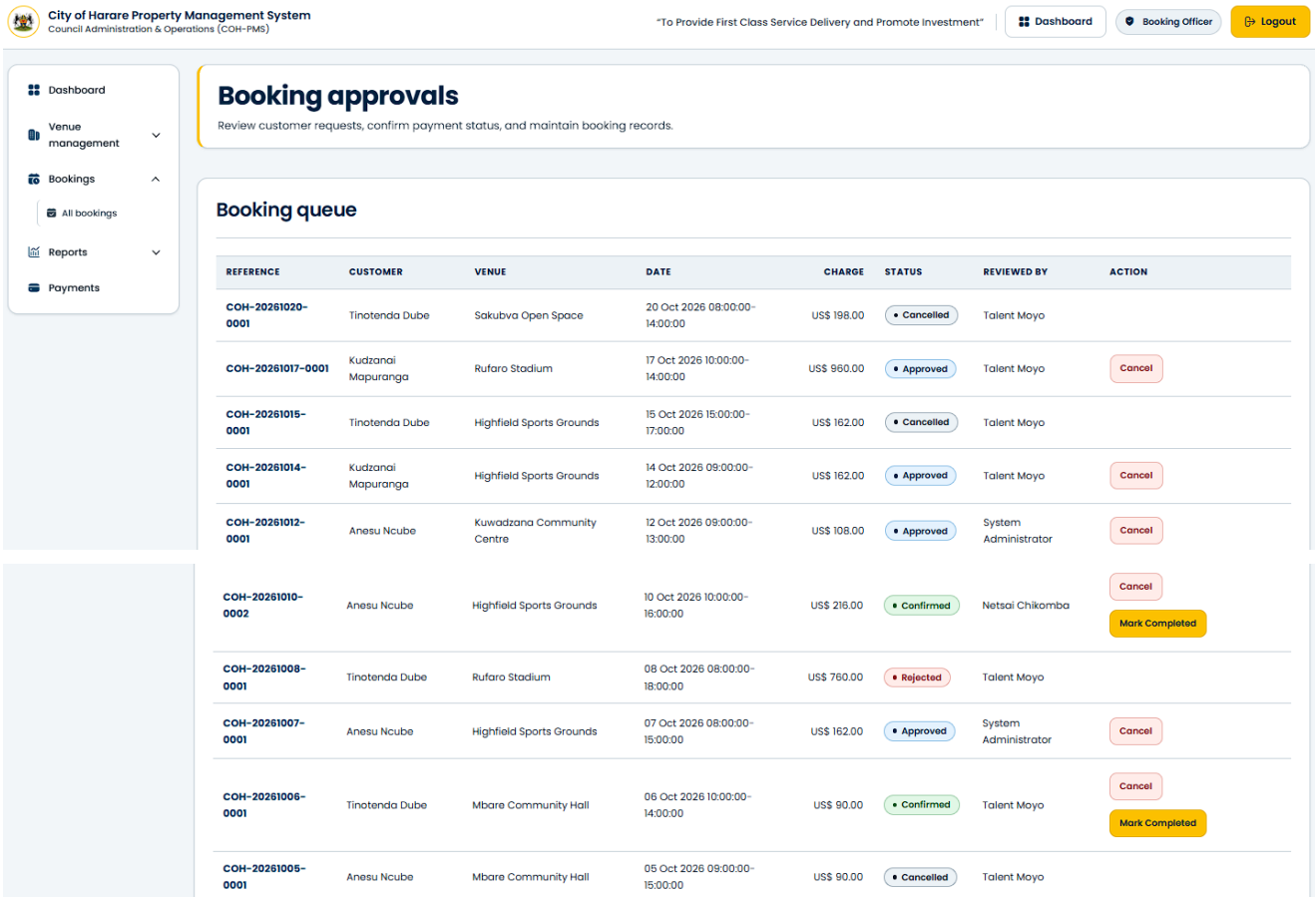
Customer Interface:
The Customer interface allows registered customers to view available venues, select a venue, choose a booking date and time, enter event details, and submit a booking request. After a booking is approved, the customer can view the booking status and submit payment details for Revenue Officer verification.



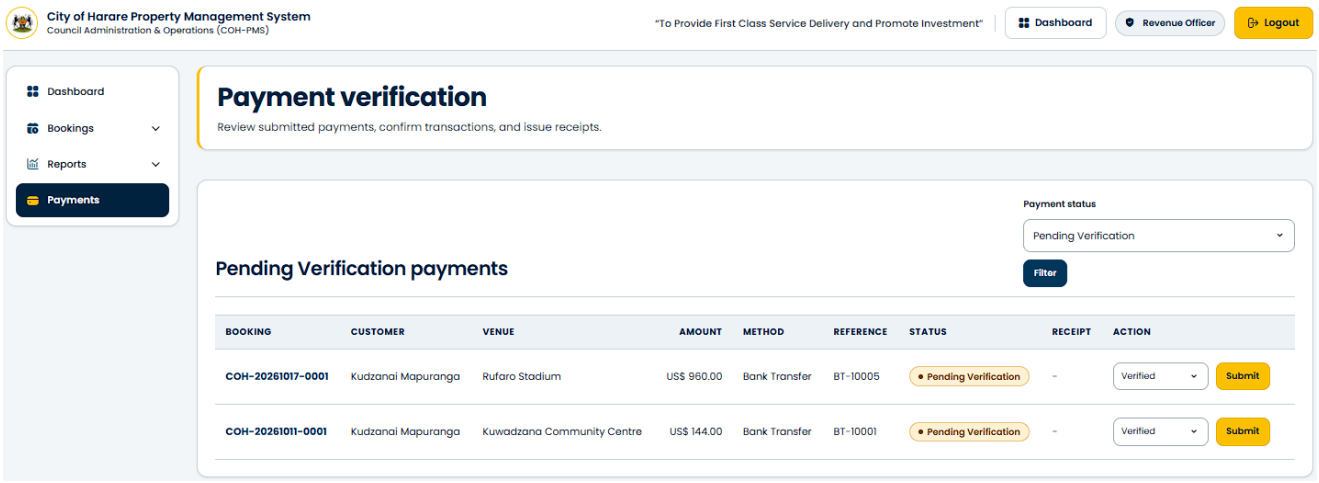


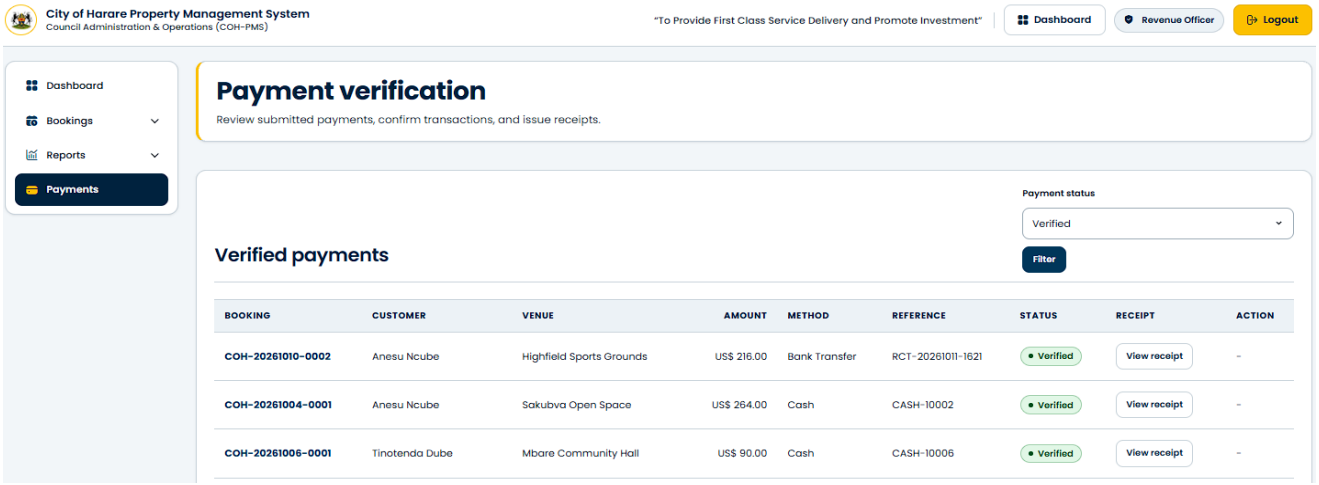


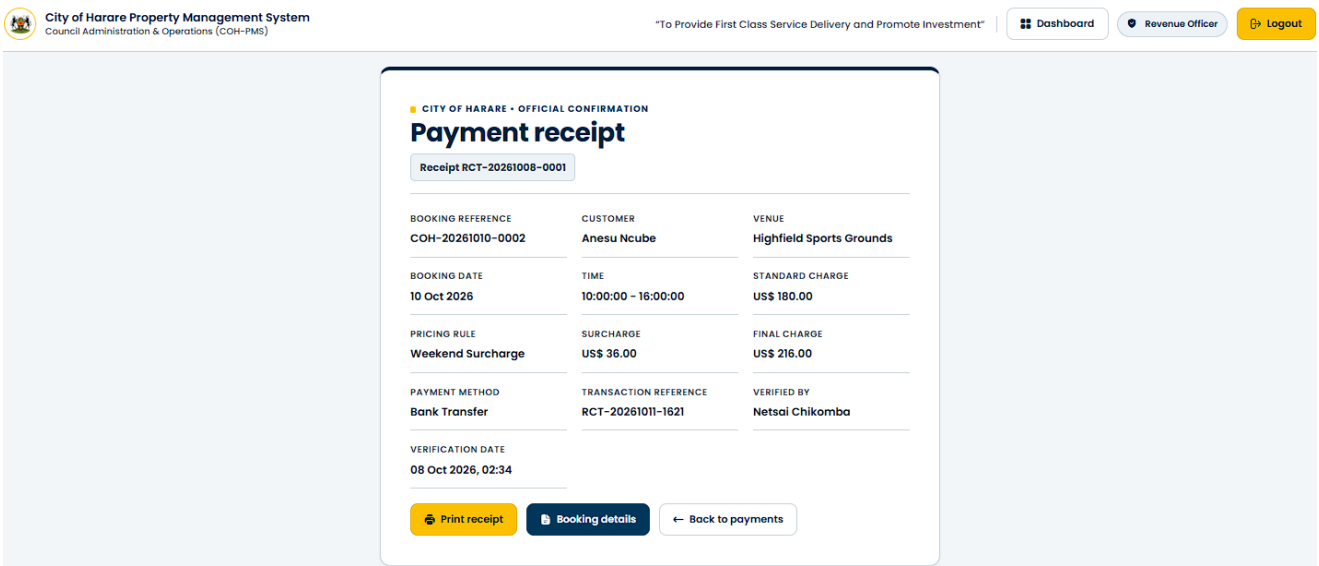
Booking Officer Interface:
The Booking Officer interface displays booking requests submitted by customers. The Booking Officer reviews the booking details, checks venue availability and capacity, and approves, reject, or cancel eligible bookings.



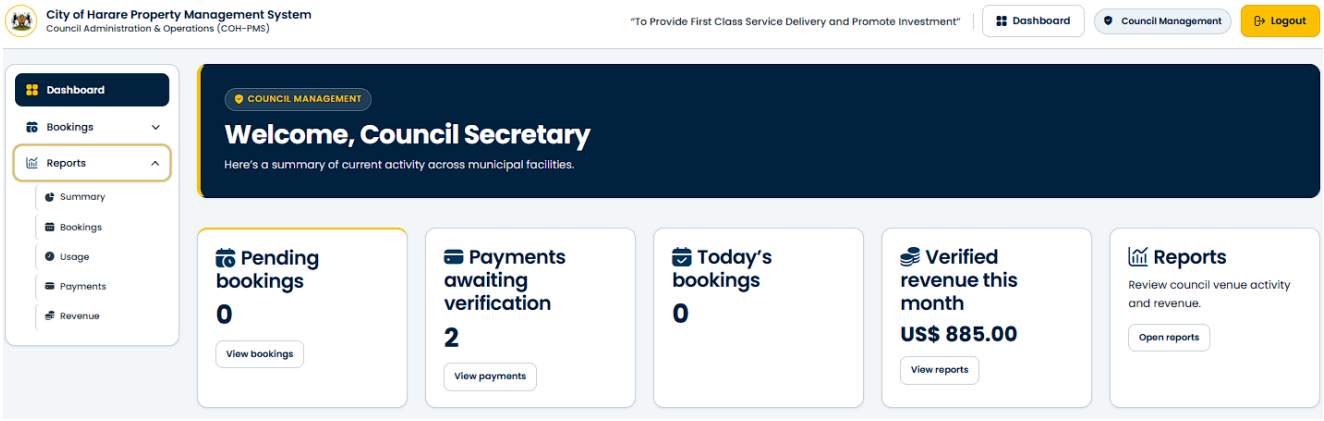
Revenue Officer Interface:
The Revenue Officer interface displays payments submitted by customers and awaiting verification. The Revenue Officer reviews the payment details, verify the payment, generate a receipt number, and confirm the related approved booking.

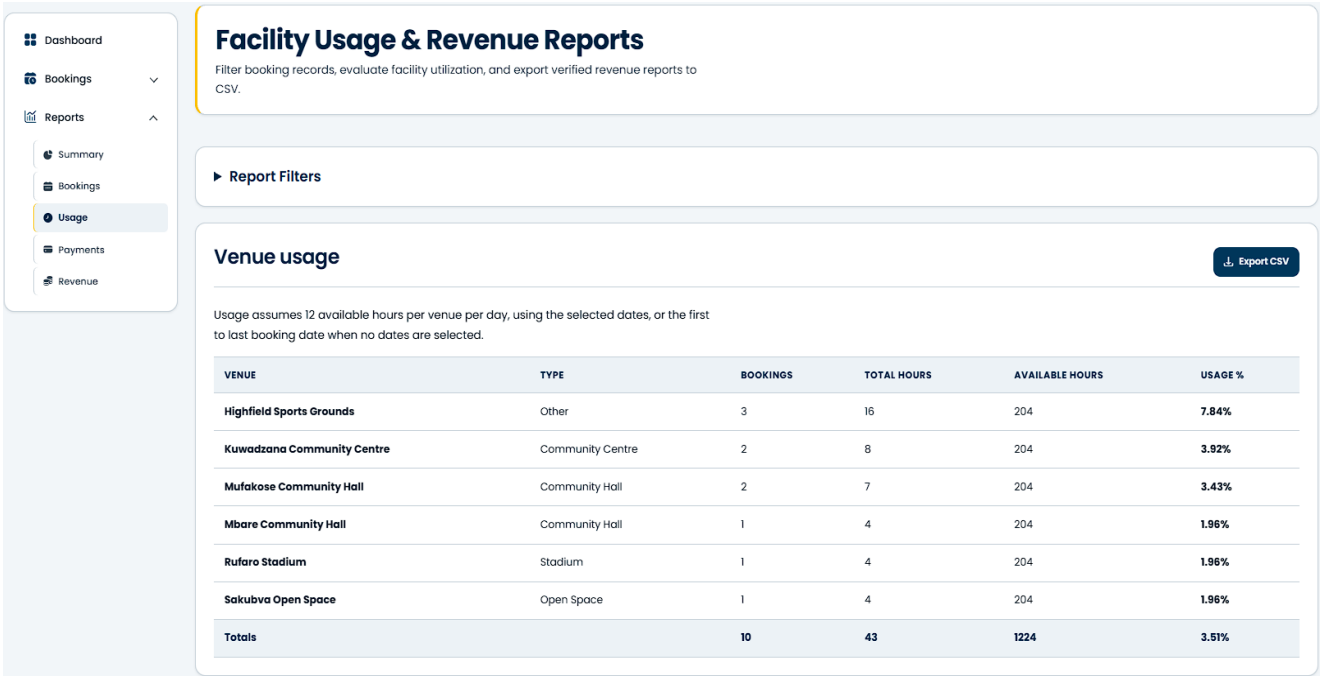


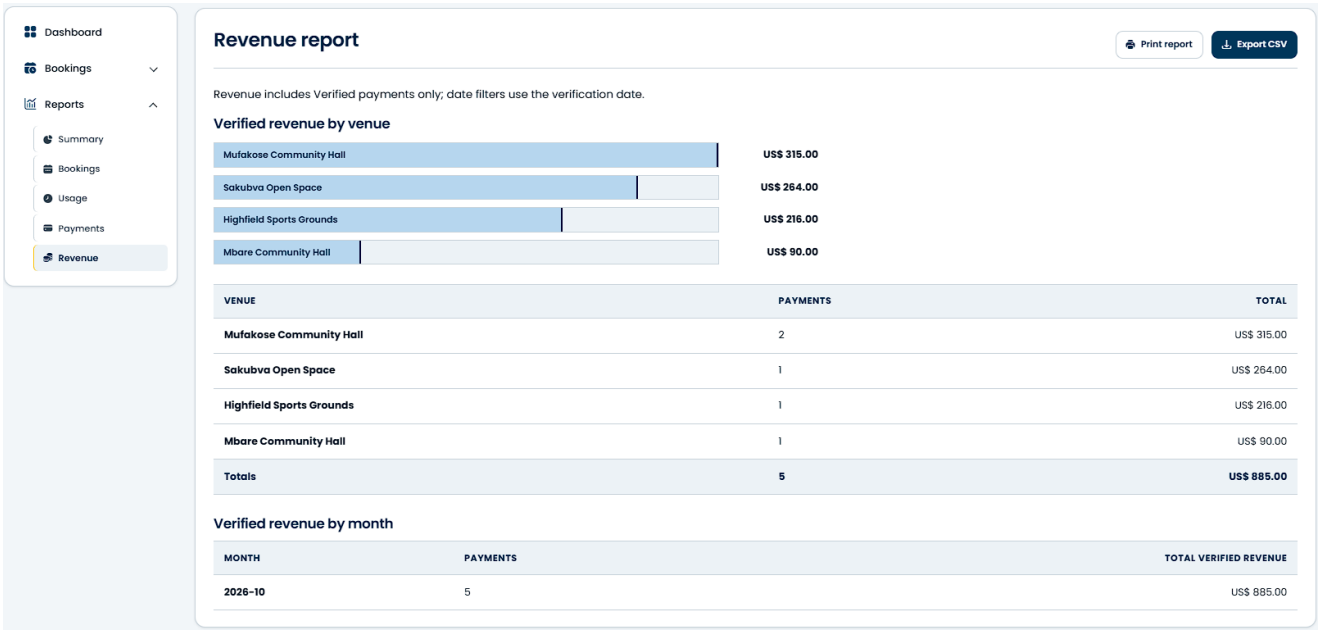




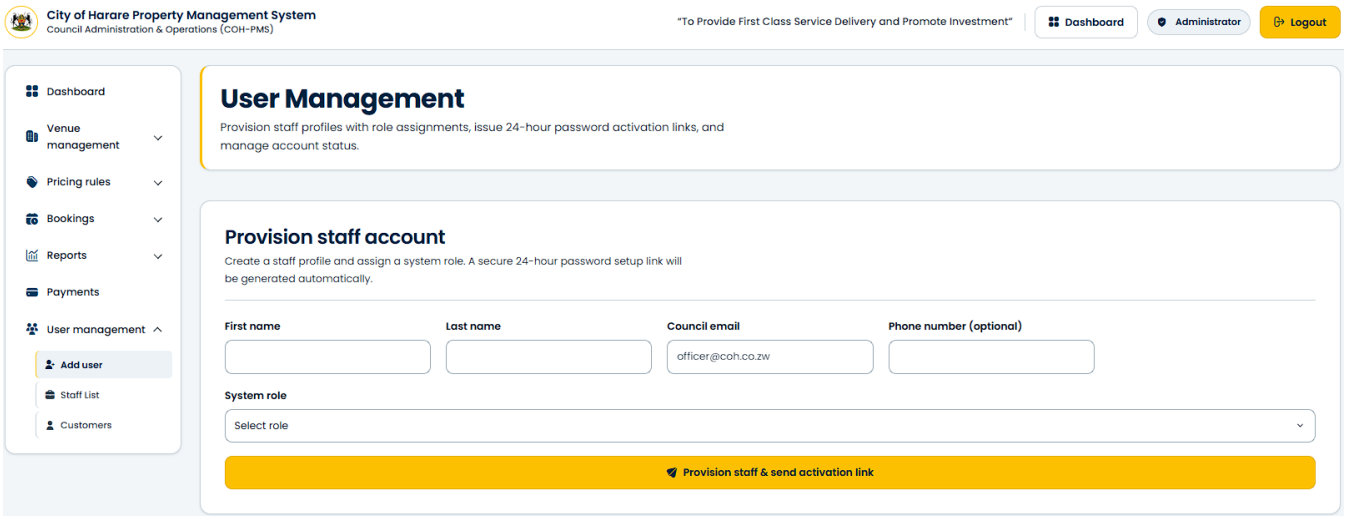
Council Management Interface:
The Council Management interface provides read-only access to system dashboards and reports. It displays side bar with booking summaries, facility-usage information, payment records, revenue performance, and monthly revenue trends to support management decision-making.

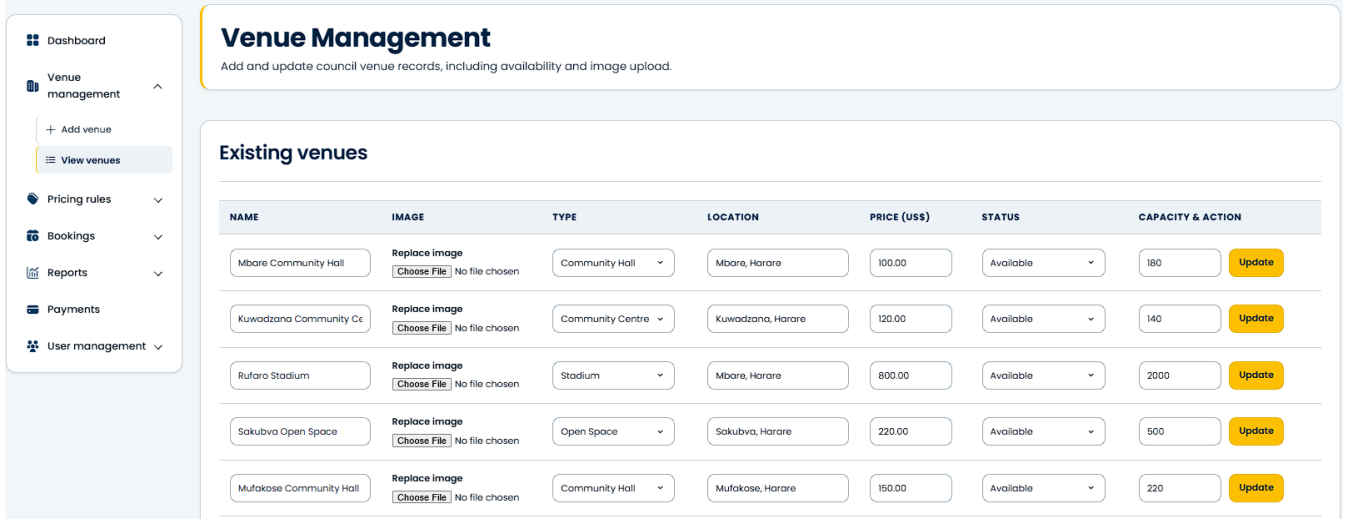


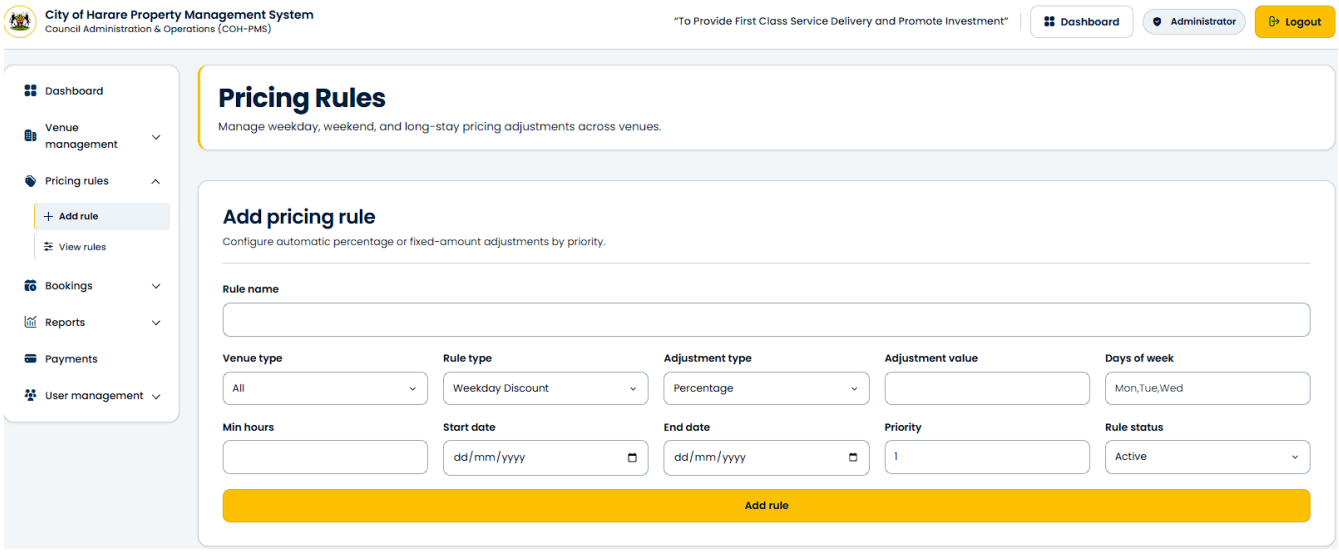




Administrator Interface:
The Administrator interface allows the System Administrator to manage user accounts, venue records, venue availability, and approved pricing rules. It also provides access to administrative functions required to maintain the system.







6.3 User Training

User training is conducted to ensure that authorised users understands how to use the COH-PMS correctly after deployment. Training focuses on the functions available to each user role and the procedures for booking venues, managing bookings, verifying payments, and accessing reports. Training ensures that each user understands their responsibilities and uses only the functions assigned to their role. It also reduces the risk of incorrect booking, payment, or reporting procedures during the changeover from the manual system to the COH-PMS.

- Customers are trained on how to register, log in, search for available venues, submit booking requests, track booking status, and submit payment details.
- Booking Officers are trained on how to review booking requests and approve, reject, or cancel bookings.
- Revenue Officers are trained on how to review submitted payments, verify payments, issue receipt numbers, and confirm approved bookings.
- Council Management users are trained on how to access dashboards and reports.
- Administrators are trained on how to manage user accounts, venues, and pricing rules.

6.4 Changeover Strategy

The COH-PMS uses a phased changeover strategy to move the City of Harare from the manual venue booking process to the new system. During the initial phase, the manual booking register and the COH-PMS are used together to allow staff to compare records, confirm booking details, and identify any operational issues.

After the system is confirmed to be working correctly, the manual booking process is gradually reduced and the COH-PMS becomes the main system for recording venue bookings, managing payments, and generating reports. This approach reduces the risk of data loss, booking errors, and disruption to Council operations during the transition.

The changeover process includes transferring existing venue and pricing information into the system, creating user accounts, training users, and monitoring system performance. Once staff and customers are familiar with the system, the manual process is discontinued and the COH-PMS is used as the official system for property and venue hire management.

6.5 Post-Implementation Support

Post-implementation support is provided to ensure that the COH-PMS continues to operate correctly after deployment. Support includes monitoring system performance, resolving user problems, correcting errors, updating venue and pricing information, and maintaining the MySQL database.

- The System Administrator is responsible for managing user accounts, reviewing system access, and performing regular database backups.
- Technical support is also provided to Customers, Booking Officers, Revenue Officers, and Council Management users when they experience difficulties using the system.
- Regular system reviews are conducted to identify areas for improvement, such as additional reports, improved payment options, or updated pricing rules.
- These activities ensure that the COH-PMS remains reliable, secure, and suitable for the on-going property and venue hire needs of the City of Harare.

6.6 Conclusion

This chapter presents the implementation of the tested City of Harare Property Management System (COH-PMS). It explained how the system is deployed, demonstrated the main system interfaces for different user roles, and outlined user training, changeover, and post-implementation support. The implementation process ensures that authorised users can access and use the system correctly in their daily work. Through phased changeover, user training, and on-going support, the COH-PMS supports the transition from the manual venue booking process to a reliable web-based property and venue hire management system.

CHAPTER VII: SUMMARY RECOMMENDATIONS AND CONCLUSION

7.1 Summary

This research investigated the challenges faced by the City of Harare in managing public properties and venue bookings using manual processes. The study proposed the development of the City of Harare Property Management System (COH-PMS) to improve venue management, automate approved pricing rules, and provide reports on facility usage and revenue.

Chapter 1 introduced the background of the study, the problem statement, system objectives, justification, and limitations. Chapter 2 reviewed literature related to property management systems, venue booking, automated pricing, and reporting.

Chapter 3 presented the requirements analysis of the COH-PMS. It identified the users of the system, their needs, and the functional and non-functional requirements. Chapter 4 presented the system design, including the system architecture, database design, user interface design, and security design.

Chapter 5 presented the coding and testing of the system. It covered the programming tools and languages used, the source code structure, key system functions, and unit, module, system, acceptance, and security testing. The test results confirmed that the system functions correctly and meets the specified requirements.

Chapter 6 presented the implementation of the COH-PMS. It explained how the system is deployed, demonstrated the main system interfaces, and described user training, changeover, and post-implementation support. The system supports Customers, Booking Officers, Revenue Officers, Council Management, and Administrators in managing venue bookings, payments, pricing rules, and reports.

7.2 Recommendations

- The City of Harare should adopt the COH-PMS as the official system for managing public properties and venue bookings.
- User training should be provided to Customers, Booking Officers, Revenue Officers, Council Management, and Administrators.
- Regular database backups should be performed to protect booking, payment, venue, and user records.
- The System Administrator should monitor the system, manage user accounts, and resolve technical problems.
- Venue information and approved pricing rules should be reviewed and updated regularly.
- The City of Harare should review the system after implementation to identify areas that require improvement.
- Future improvements should include mobile application support, online payment integration, and automated booking and payment notifications.
- Additional reports should be developed to support management decision-making.

7.3 Recommendations for Further Research

- Online payment integration should be considered to allow customers to pay booking charges through secure payment platforms.
- Automated email and SMS notifications should be added to inform customers about booking approval, payment verification, and booking cancellation.
- The system should be expanded to include property maintenance records and other Council services.
- Advanced reports should be developed to analyse customer demand, venue performance, seasonal booking trends, and revenue collection.
- The system should be integrated with other City of Harare systems to improve information sharing and service delivery.

7.4 Conclusion

The COH-PMS provides a practical solution to the challenges of manual property and venue booking management. The system improves venue management, booking processing, payment recording, and reporting. The system applies approved pricing rules and provides facility-usage and revenue reports for management decision-making. The successful implementation of the COH-PMS will reduce manual paperwork, improve record keeping, and support efficient service delivery. The City of Harare should continue maintaining and improving the system to meet future property and venue hire needs.

CHAPTER VIII: BIBLIOGRAPHY

Booch, G., Rumbaugh, J. and Jacobson, I. 2005. *The Unified Modeling Language User Guide*. 2nd ed. Upper Saddle River: Addison-Wesley.

City of Harare. 2026. *City of Harare Official Website*. Available at: City of Harare Portal [Accessed: 02 October 2026].

Connolly, T.M. and Begg, C.E. 2005. *Database Systems: A Practical Approach to Design, Implementation, and Management*. 4th ed. Harlow: Addison-Wesley.

Fjeldstad, O.-H. and Heggstad, K.K. 2012. *Local Government Revenue Mobilisation in Anglophone Africa*. CMI Working Paper WP 2012:6. Bergen: Chr. Michelsen Institute.

Fowler, M. 2002. *Patterns of Enterprise Application Architecture*. Boston: Addison-Wesley.

Government of Zimbabwe. 2021. *Cyber and Data Protection Act [Chapter 12:07] (Act No. 5 of 2021)*. Harare: Government Printer. Available at: ZimLII Act Portal [Accessed: 04 October 2026].

Government of Zimbabwe. 2025. *National Development Strategy 2 (NDS2): January 2026 to December 2030*. Harare: Government of Zimbabwe.

Kendall, K.E. and Kendall, J.E. 2020. *Systems Analysis and Design*. 10th ed., Global ed. Harlow: Pearson.

Larman, C. 2004. *Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development*. 3rd ed. Upper Saddle River: Prentice Hall.

McCluskey, W., Franzsen, R., Kabinga, M. and Kasese, C. 2018. *The Role of Information Communication Technology to Enhance Property Tax Revenue in Africa: A Tale of Four Cities in Three Countries*. ICTD Working Paper 88. Brighton: Institute of Development Studies.

Mozilla and individual contributors. 2026. *MDN Web Docs*. Available at: MDN Web Docs Platform [Accessed: 05 October 2026].

Myers, G.J., Sandler, C. and Badgett, T. 2011. *The Art of Software Testing*. 3rd ed. Hoboken: John Wiley & Sons.

Nielsen, J. 1994. *10 Usability Heuristics for User Interface Design*. Nielsen Norman Group. Available at: NNG Heuristics Resource [Accessed: 28 September 2026].

Oracle Corporation. 2026. *MySQL 8.4 Reference Manual*. Available at: MySQL Developer Documentation [Accessed: 01 October 2026].

OWASP Foundation. 2025. *OWASP Top 10:2025*. Available at: OWASP Top 10 Security Project [Accessed: 18 September 2026].

The PHP Group. 2026. *PHP Manual*. Available at: Official PHP Manual [Accessed: 08 October 2026].

Pressman, R.S. and Maxim, B.R. 2020. *Software Engineering: A Practitioner's Approach*. 9th ed. New York: McGraw-Hill Education.

Sommerville, I. 2016. *Software Engineering*. 10th ed., Global ed. Harlow: Pearson.

Stuttard, D. and Pinto, M. 2011. *The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws*. 2nd ed. Indianapolis: John Wiley & Sons.

W3C. 2024. *Web Content Accessibility Guidelines (WCAG) 2.2*. Available at: W3C Standards Directory [Accessed: 25 September 2026].

Welling, L. and Thomson, L. 2016. *PHP and MySQL Web Development*. 5th ed. Boston: Addison-Wesley.

APPENDIX A: RESEARCH QUESTIONNAIRE

City of Harare Property Management System (COH-PMS)
City of Harare Venue Hire | Kushinga Phikelela Polytechnic | Academic Year: 2026

INSTRUCTIONS: This questionnaire is intended to gather information about how council venues (community halls, community centres, stadiums and open spaces) are currently booked, priced, paid for and reported on in the City of Harare. Taking part is voluntary and your answers are anonymous. Please do not write your name. All information collected will be used strictly for academic research purposes. Please answer each question by ticking the option that best represents your experience. Questions marked (Council staff only) can be skipped by customers.

SECTION A: RESPONDENT INFORMATION

1. What is your role?

- Customer (hirer of council venues)
- Booking clerk or booking officer
- Revenue or cashier officer
- Council management or supervisor
- Other (please specify):_____

2. How long have you been involved in council venue hire, as a customer or as staff?

- Less than 1 year
- 1-3 years
- 4-6 years
- More than 6 years

3. How often do you hire, or process the hire of, council venues?

- Very often (every week)
- Often (about once a month)
- Occasionally (a few times a year)
- Rarely (once a year or less)

SECTION B: CURRENT BOOKING PRACTICES

4. How is venue bookings currently made? (Tick all that apply)

- Visiting the council offices in person
- Telephone call
- Paper booking register
- Through an agent or another person
- Other (please specify): _____

5. How long does it usually take to have a booking confirmed, from the first enquiry?

- The same day
- 1-3 days
- 4-7 days
- More than a week

6. How easy is it to find out whether a venue is free on a particular date and time?

- Very easy
- Easy
- Difficult
- Very difficult

7. How often have you experienced, or heard of, double bookings or date clashes?

- Very often
- Often
- Occasionally
- Never

SECTION C: PRICING, PAYMENTS AND RECEIPTS

8. How do you find out the cost of hiring a venue before you book? (Tick all that apply)

- Asking at the council offices
- Printed price list or tariff
- Telephone enquiry
- Word of mouth
- The cost is only known when payment is made

9. Have you ever been charged a price different from what you expected, or different from what another customer paid for similar hire?

- Yes, often
- Yes, sometimes
- No
- Not sure

10. How is hire fees usually paid? (Tick all that apply)

- Cash at the council offices
- Bank deposit or transfer
- Mobile money (for example EcoCash)
- Other (please specify):_____

11. How long does it take to receive an official receipt or proof of payment?

- Immediately
- The same day
- 1-3 days
- More than 3 days

12. (Council staff only) How are payments currently recorded and checked?

- Paper receipt books
- Spreadsheet
- Accounting or other software
- Other (please specify):_____

SECTION D: RECORDS AND REPORTING (Council staff only)

13. How is venue usage and revenue reports currently produced?

- Manually from paper records
- From spreadsheets
- From an existing computer system
- They are not produced regularly

14. How long does it take to prepare a monthly venue usage and revenue report?

- The same day
- 1-3 days
- About a week
- More than a week

15. How confident are you in the accuracy of the current booking and revenue records?

- Very confident
- Confident
- Neutral
- Not confident

SECTION E: TECHNOLOGY READINESS

16. Which devices do you have access to? (Tick all that apply)

- Smartphone
- Laptop or desktop computer
- Tablet
- None

17. How reliable is your internet access?

- Yes, always
- Sometimes
- Rarely
- Never

18. How comfortable are you using online services such as email, online forms or mobile money?

- Very comfortable
- Comfortable
- Neutral
- Not comfortable

19. Would you use an online system to check venue availability, see the price, book and record payment?

- Definitely
- Probably
- Not sure
- No

20. Which features would matter most to you in such a system? (Tick up to three)

- Checking venue availability online
- Seeing the full price before booking
- Submitting a booking request online
- Tracking the status of my booking
- Recording payment and receiving a receipt
- Management reports on venue usage and revenue (staff)

Thank you for your time and participation in this research.

APPENDIX B: OBSERVATION CHECKLIST

Observation of the current venue hire process | Kushinga Phikelela Polytechnic | Academic Year 2026

Date: _____ Time: _____ Location: _____ Observer:_____

Observation Area	Observed (Yes/No)	Rating (1–5)	Comments
Booking enquiry and handling of customer requests			
Checking venue availability (register, phone, memory)			
Evidence of double bookings or date clashes			
Display and use of venue price lists or tariffs			
Consistency of prices charged for similar bookings			
Application of discounts or surcharges			
Booking approval process			
Payment collection and verification process			
Issuing of receipts			
Record keeping of bookings and payments			
Preparation of usage and revenue reports			
Customer waiting time and queues at the office			
Availability and use of computers or smartphones in the office			

Rating Scale:
1 = None/Not Observed
2 = Minimal
3 = Moderate
4 = Significant
5 = Severe/Constant

For each area, the rating shows how strongly the weakness or manual dependence was observed.

APPENDIX C: INTERVIEW GUIDE

Semi-Structured Interview Guide: City of Harare Booking Officer and Council Staff

Interviewee role:_____ Date:_____

Explain to the interviewee that taking part is voluntary, that their name will not be recorded, and that the answers will be used only for academic research.

PART A: CURRENT VENUE BOOKING PROCESS

1. How is a venue booked at the moment, from the first enquiry to the final confirmation?
2. How do you check whether a venue is free on the date and time a customer asks for?
3. How often do double bookings or date clashes happen, and how are they resolved?

PART B: PRICING AND PAYMENTS

4. How are hire charges decided, and who is responsible for setting or changing them?
5. Are discounts or surcharges applied, for example for weekdays, weekends or long bookings? How are they calculated and recorded?
6. How are payments received and checked, and how are official receipts issued?
7. What problems do you experience with payments, such as late payment, wrong amounts or payments that cannot be verified?

PART C: RECORDS AND REPORTING

8. How are bookings and payments recorded, and where are the records kept?
9. What reports does management ask for, how long do they take to prepare, and what errors or delays occur?

PART D: SYSTEM REQUIREMENTS

10. What information must be recorded for every booking?
11. Which staff should be allowed to approve bookings, verify payments, change prices and view reports?
12. Would a web-based venue booking system be useful, and what could make it difficult to adopt, for example training, devices or internet access?
13. What features would be most important to you in such a system?

Thank you for your valuable contribution to this research.